

# Elbow and forearm

## SKIN AND SOFT TISSUES

### SKIN

#### Cutaneous vascular supply

The skin of the elbow and forearm receives its blood supply from muscle perforators, fasciocutaneous networks and direct cutaneous vessels. The skin of the anterolateral cubital fossa is supplied by muscle perforators arising from the radial collateral and radial recurrent arteries. The skin of the medial cubital fossa receives branches from an anastomosis between the inferior ulnar collateral and the ulnar recurrent arteries. Small direct cutaneous vessels spring from the brachial artery. The skin of the posterolateral aspect of the elbow is supplied by branches of the posterior interosseous recurrent artery and a rich plexus of collateral and anastomosing arteries. The skin of the posteromedial aspect is supplied by branches of the superior and inferior ulnar collateral arteries and the posterior ulnar recurrent artery feeding into the plexus.

The skin of the anterior forearm is supplied by fasciocutaneous perforators; the largest is the inferior cubital artery, which arises from the radial artery at the distal apex of the cubital fossa and extends to midway down the forearm. Fasciocutaneous perforators from the radial artery supply the lateral two-thirds of the anterior surface of the forearm (see Fig. 46.5). The vessels pass in the intermuscular septa between brachioradialis and flexor carpi radialis, and between flexor carpi radialis and flexor digitorum superficialis. Fasciocutaneous perforators from the ulnar artery supply skin over the ulnar side of the anterior forearm and the medial quarter of the posterior forearm from the cubital fossa to the wrist. A small contribution comes from musculocutaneous perforators that pass through brachioradialis and flexor carpi ulnaris.

The skin of the distal third of the lateral forearm is supplied by the terminal perforating branches of the anterior interosseous artery. The central half of the posterior surface of the forearm, from the lateral edge of extensor digitorum, just below the lateral epicondyle of the humerus, to the wrist, is supplied by fasciocutaneous perforators from the posterior interosseous artery that pass along the intermuscular fascia between extensor carpi ulnaris and extensor digiti minimi.

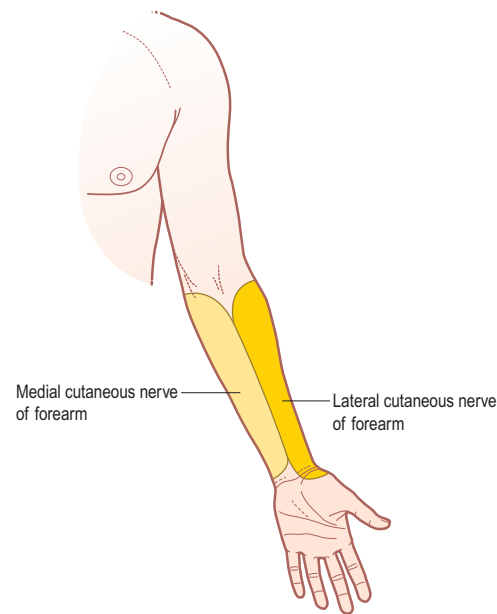
#### Cutaneous innervation

A small area of skin anterolateral to the elbow is supplied by the lateral cutaneous nerve of the arm. The skin overlying the posterior aspect of the elbow is innervated by the posterior cutaneous nerve of the arm and proximal branches of the posterior cutaneous nerve of the forearm. The lateral cutaneous nerve of the forearm innervates the skin of the anterior half and radial posterior quarter of the forearm. The medial cutaneous nerve of the forearm supplies the skin overlying most of the elbow, the ulnar half of the anterior forearm and the ulnar quarter of the posterior forearm (Fig. 49.1 and p. 860).

### SOFT TISSUES

#### Cubital fossa

The cubital fossa is a triangular depression in the anterior aspect of the upper forearm (see Fig. 46.29). It is bounded proximally by the epicondylar line, medially by the lateral margin of pronator teres, and laterally by the medial face of brachioradialis; it is bisected by the tendon of biceps brachii. The deep fascia of the forearm forms its roof. It is continuous with the deep fascia of the anterior arm, attached to



**Fig. 49.1** Sensory loss or disturbance after complete section of the medial and lateral cutaneous nerves of the forearm.

the medial and lateral humeral epicondyles and augmented by the bicipital aponeurosis, which sweeps across the medial and inferior part of the fossa to the fascia over the superficial flexor muscles. Supinator and brachialis lie in the floor of the fossa. From medial to lateral, the basilic, median cubital and cephalic veins lie in the superficial fascia. Branches of the medial cutaneous nerve of the forearm are closely related to the basilic vein. The lateral cutaneous nerve of the forearm runs down alongside the lateral edge of the tendon of biceps brachii. Deep to the deep fascia, and passing from medial to lateral, are the median nerve and the brachial artery, the latter inclining from the medial side of the arm to pass down the fossa close to the midline. The radial and ulnar arteries, with their accompanying veins, originate in the lower part of the fossa. The radial nerve enters the fossa laterally in the groove between brachioradialis and brachialis, and divides into the superficial radial and posterior interosseous nerves above the humero-radial joint. The superficial radial nerve continues down the forearm under cover of brachioradialis.

#### Compartments of the forearm

The deep fascia, the interosseous membrane and the fibrous intermuscular septa, some of which pass between the deep fascia and bone, divide the forearm into compartments. These include the superficial and deep flexor (anterior) compartments, the extensor (posterior) compartment, and a proximolateral compartment that includes brachioradialis and extensors carpi radialis longus and brevis (the mobile wad) (Fig. 49.2). The dense sheath of the antebrachial fascia (the deep fascia of the forearm) is continuous above with the brachial fascia. It is attached to the olecranon, the posterior border of the ulna, and the lateral and medial epicondyles. The fascia is thicker posteriorly and in the lower forearm. It is strengthened above by tendinous fibres from biceps and triceps. Near the wrist, two localized thickenings, the flexor and extensor retinacula, retain the digital tendons in position (Ch. 50). From its deep surface, septa pass between muscles, providing areas of additional attachment. Muscles arise from its inner aspect in the upper forearm. Vessels and nerves pass through apertures in the fascia; a large aperture anterior to the elbow transmits a venous communication between superficial and deep veins.

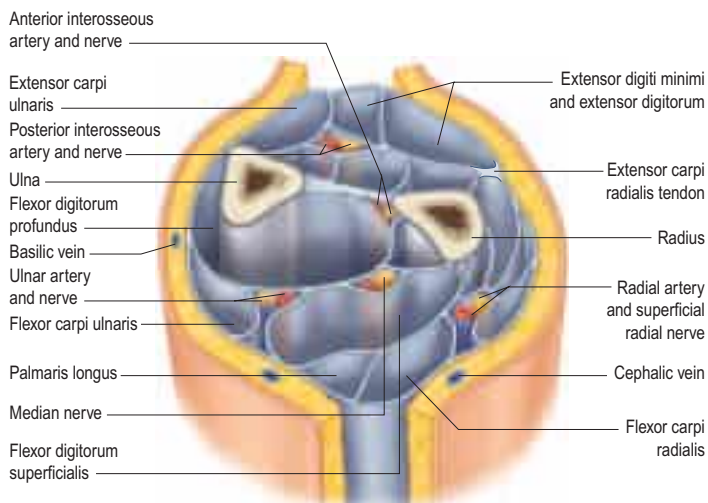
## BONES

## HUMERUS

**Distal end of humerus**

The distal end of the humerus is a modified condyle; it is wider transversely and has articular and non-articular parts (Figs 49.3–49.4). The articular part is curved forwards, so that its anterior and posterior surfaces lie in front of the corresponding surfaces of the shaft. It articulates with the radius and the ulna at the elbow joint, and is divided by a faint groove into a lateral capitulum and a medial trochlea.

The capitulum is a rounded, convex projection, considerably less than half a sphere, which covers the anterior and inferior surfaces of the lateral part of the condyle of the humerus but does not extend on to its posterior surface. It articulates with the discoid head of the radius, which lies in contact with its inferior surface in full extension of the elbow but slides on to its anterior surface during flexion. The groove of the trochlea winds backwards and laterally from the anterior to the posterior surface of the bone, and is wider, deeper and more symmetrical posteriorly. Anteriorly, the medial flange of the pulley is longer than the lateral, and the surface adjoining its projecting medial margin is convex to accommodate itself to the medial part of the upper surface of the coronoid process of the ulna. These asymmetries entail varying angulation between the humeral and ulnar axes, together with some conjunct rotation. The non-articular part of the condyle includes the medial and lateral epicondyles, and the olecranon, coronoid and radial fossae.



**Fig. 49.2** Compartments of the forearm. A cross-section through the upper third of the left forearm.

**Trochlea**

The trochlea is a pulley-shaped surface that covers the anterior, inferior and posterior surfaces of the condyle of the humerus medially and articulates with the trochlear notch of the ulna. On its lateral side, it is separated from the capitulum by a faint groove; its medial margin projects distally beyond the rest of the bone. When the elbow is extended, the inferior and posterior aspects of the trochlea are in contact with the ulna but, as the joint is flexed, the trochlear notch slides on to the anterior aspect so that its posterior aspect is then left uncovered. The downward projection of the medial edge of the trochlea is the principal factor in determining the degree of angulation between the long axis of the humerus and the long axis of the supinated forearm when the elbow is extended.

**Medial epicondyle**

The medial border of the humerus ends by turning slightly backwards as the medial epicondyle, which forms a conspicuous, blunt projection on the medial side of the condyle. It is subcutaneous and usually visible, especially in passive flexion of the elbow. Its posterior surface is smooth and is crossed by the ulnar nerve, which lies in a shallow sulcus as it enters the forearm. The nerve can be felt and rolled against the bone here; if jarred against the epicondyle, a characteristic tingling sensation results. The lower part of the anterior surface of the medial epicondyle is marked by the attachment of the superficial group of forearm flexors. They arise from the epiphysis for the epicondyle but are entirely extracapsular.

**Lateral epicondyle**

The lateral border of the humerus ends at the lateral epicondyle; its lower portion constitutes the lateral supracondylar ridge. The lateral epicondyle occupies the lateral part of the non-articular portion of the condyle but does not project beyond the lateral supracondylar ridge. It



**Fig. 49.4** The inferior aspect of the distal end of the left humerus. Key: 1, trochlea; 2, capitulum; 3, medial epicondyle; 4, sulcus for ulnar nerve; 5, lateral epicondyle.



**Fig. 49.3 A**, The anterior aspect of the distal end of the left humerus. Key: 1, shaft; 2, medial supracondylar ridge; 3, coronoid fossa; 4, medial epicondyle; 5, trochlea; 6, lateral supracondylar ridge; 7, radial fossa; 8, lateral epicondyle; 9, capitulum. **B**, The posterior aspect of the distal end of the left humerus. Key: 1, lateral supracondylar ridge; 2, lateral epicondyle; 3, medial supracondylar ridge; 4, medial epicondyle; 5, olecranon fossa; 6, sulcus for ulnar nerve; 7, trochlea.

turns slightly forwards, unlike the medial epicondyle, which turns slightly backwards. Its lateral and anterior surfaces show a well-marked impression for the superficial group of the extensor muscles of the forearm (see Fig. 48.11A), which arise from the lateral side of the lower humeral epiphysis; like the flexors, they are extracapsular. The posterior surface, which is very slightly convex, is easily felt in a depression visible behind the extended elbow. A small area on the posterior surface gives origin to anconeus.

### **Olecranon, coronoid and radial fossae**

The olecranon fossa is a deep hollow on the posterior surface of the condyle, immediately above the trochlea, that lodges the tip of the olecranon of the ulna when the elbow is extended. The floor of the fossa is always thin and may be partially deficient. A similar but smaller hollow, the coronoid fossa, lies immediately above the trochlea on the anterior surface of the condyle and accommodates the anterior margin of the coronoid process of the ulna during flexion of the elbow. A very slight depression, the radial fossa, lies above the capitulum on the lateral side of the coronoid fossa and is related to the margin of the head of the radius in full flexion of the elbow.

### **Ossification**

The distal humerus ossifies as follows. During the first year, ossification begins in the capitulum and extends medially to form the main part of the articular surface. The centre for the medial part of the trochlea appears in the ninth year in females and tenth year in males. Ossification begins in the medial epicondyle in the fourth year in females and the sixth in males, and in the lateral epicondyle in about the twelfth year. The centre for the medial epicondyle forms a separate epiphysis, which is entirely extracapsular, on the posteromedial aspect of the epicondyle. It is separated from the rest of the lower epiphysis by a downgrowth from the shaft, with which it unites at about the twentieth year. The centres for the lateral epicondyle, capitulum and trochlea fuse around puberty, and the composite epiphysis unites with the shaft in the fourteenth year in females and sixteenth in males.

## **RADIUS**

The radius is the lateral bone of the forearm (Figs 49.5–49.7). It has expanded proximal and distal ends; the distal is much the broader. The shaft widens rapidly towards its distal end, and is convex laterally and concave anteriorly in its distal part. The radius articulates with the humeral capitulum, the ulna and the carpus.

### **Proximal radius**

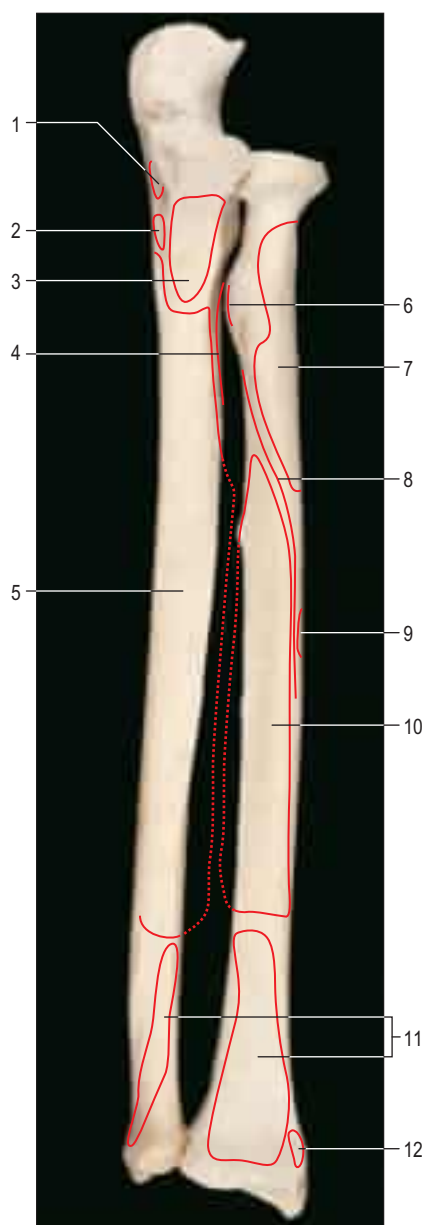
The proximal radius includes a head, neck and tuberosity. The head is discoid, its proximal surface a shallow cup for articulation with the humeral capitulum. Its smooth articular periphery is vertically deepest medially, where it contacts the ulnar radial notch. The posterior surface of the radial head is palpable on the posterolateral aspect of the extended elbow. The head overhangs the neck, which is the circumferential constriction immediately below it, particularly laterally. The tuberosity is distal to the medial part of the neck; posteriorly, it is rough, whereas anteriorly, it is usually smooth.

### **Radial shaft**

The radial shaft has a lateral convexity and is triangular in cross-section. The interosseous border is sharp, except proximally, near the tuberosity, and distally, where the interosseous border is the posterior margin of a small, elongated, triangular area, proximal to the ulnar notch. These two areas form the medial surface of the radial shaft. The interosseous membrane is attached to the distal three-quarters of the radial shaft, and connects the radius to the ulna. The anterior surface, between anterior and interosseous borders, is concave transversely and shows a distal anterior curvature. There is a proximally directed nutrient foramen and canal near the midpoint of the shaft. The posterior surface of the shaft, between the interosseous and posterior borders, is largely flat but may be slightly hollow proximally. The lateral surface is gently convex. A finely irregular oval area occurs near the midshaft; further distally, the surface is smooth.

### **Distal radius**

The distal end of the radius (Fig. 49.8) is the widest part and is four-sided in section. The lateral surface is slightly rough, projecting distally as the radial styloid process. The smooth carpal articular surface is divided by a ridge into medial and lateral areas. The medial distal radial articular surface is quadrangular, whereas the lateral part of the articular surface is triangular and curves on to the styloid process. The anterior surface of the distal radius is a thick, palpable, prominent ridge 2 cm



**Fig. 49.5** The articulated radius and ulna: anterior view, showing muscle attachments. Note that the hatched lines denote extension of muscular attachments on to the interosseous membrane. Key: 1, flexor digitorum superficialis; 2, pronator teres (ulnar head); 3, brachialis; 4, supinator; 5, flexor digitorum profundus; 6, biceps; 7, supinator; 8, flexor digitorum superficialis (oblique line); 9, pronator teres; 10, flexor pollicis longus; 11, pronator quadratus; 12, brachioradialis.

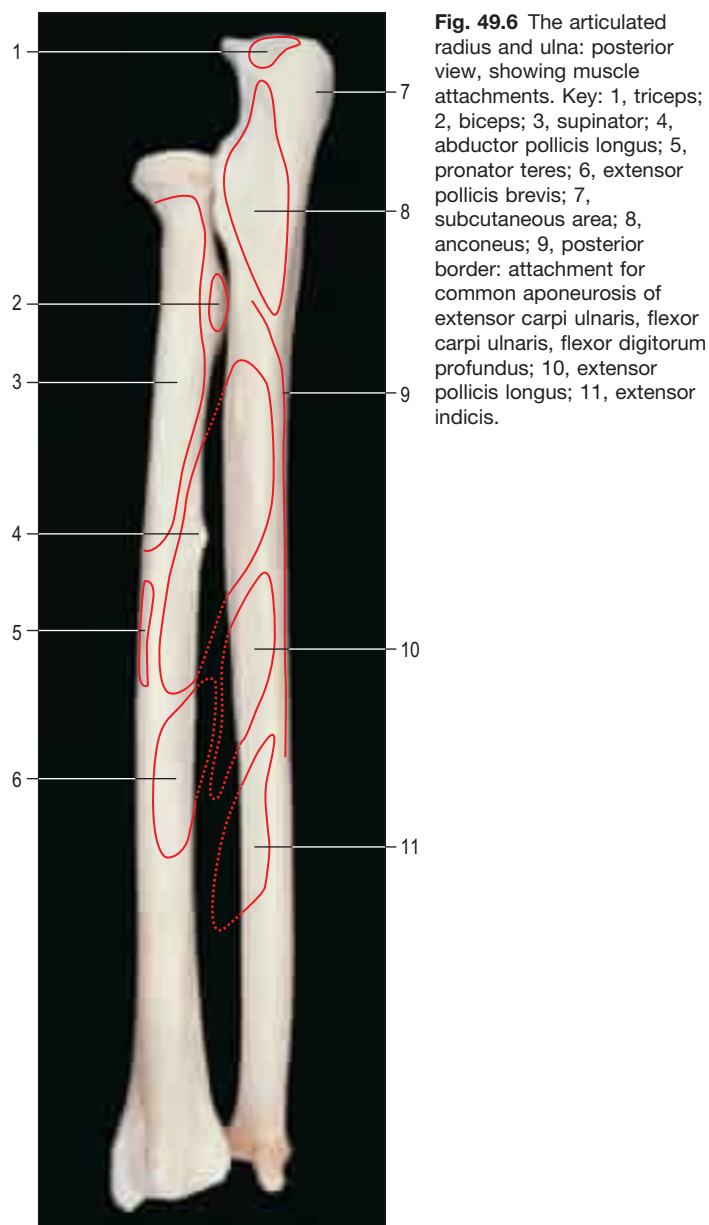
proximal to the thenar eminence. The medial surface is the ulnar notch, which is smooth and anteroposteriorly concave for articulation with the head of the ulna. The posterior surface has a palpable dorsal tubercle (Lister's tubercle), limited medially by an oblique groove and in line with the cleft between the index and middle fingers. Lateral to the tubercle is a wide, shallow groove, divided by a vertical ridge.

### **Muscular, ligamentous and articular attachments**

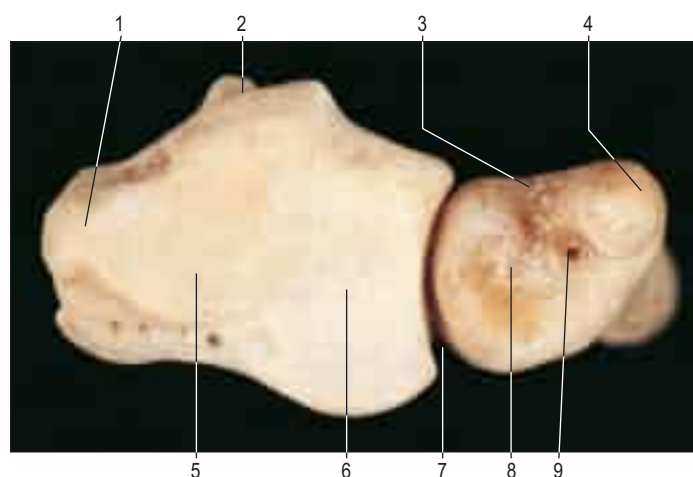
The proximal articular surface of the radial head and its circumference are covered by hyaline cartilage. The upper rim fits the groove between the capitulum and trochlea and enters the radial fossa in flexion. The articular circumference articulates with the ulnar radial notch and anular ligament, within which it rotates in pronation and supination. The radial neck is enclosed by the narrower distal part of the ligament, from which it is separated by a synovial protrusion from the superior radio-ulnar joint. The posterior area of the tuberosity is marked by the tendon of biceps brachii, which is separated from a smooth anterior area by a bursa. The oblique cord is attached just distal to the bursa.

Flexor digitorum superficialis is attached to the proximal anterior surface of the radius by a thin, wide head. Flexor pollicis longus has an extensive attachment to the proximal two-thirds of the anterior surface of the radial shaft, overlying the nutrient foramen. Pronator quadratus is attached to the distal quarter of the anterior radial shaft; a small, triangular area proximal to the ulnar notch gives attachment to the deepest part of the muscle. Pronator teres is attached to a rough area near the midpoint of the lateral surface of the radial shaft, at its maximal curvature. Proximally, supinator is attached to a long, V-shaped area on the lateral surface of the shaft. Distal to pronator teres, the lateral





**Fig. 49.6** The articulated radius and ulna: posterior view, showing muscle attachments. Key: 1, triceps; 2, biceps; 3, supinator; 4, abductor pollicis longus; 5, pronator teres; 6, extensor pollicis brevis; 7, subcutaneous area; 8, anconeus; 9, posterior border: attachment for common aponeurosis of extensor carpi ulnaris, flexor carpi ulnaris, flexor digitorum profundus; 10, extensor pollicis longus; 11, extensor indicis.



**Fig. 49.8** The distal end of the left radius and ulna. Key: 1, styloid process of radius; 2, dorsal tubercle; 3, groove for tendon of extensor carpi ulnaris; 4, styloid process of ulna; 5, facet for scaphoid; 6, facet for lunate; 7, inferior radio-ulnar articulation; 8, area of contact with articular disc; 9, area for attachment of articular disc. Compare with Fig. 50.18A.

surface is covered by tendons of brachioradialis, extensor carpi radialis longus and brevis. On the posterior surface, abductor pollicis longus is attached proximally, and extensor pollicis brevis more distally. The remaining surface is devoid of attachments and covered by the long and short extensors of the thumb. The lateral surface, proximal to the styloid process, receives the attachment of brachioradialis and is crossed obliquely, downwards and forwards, by the tendons of abductor pollicis longus and extensor pollicis brevis. The radial styloid process projects beyond that of the ulna, its apex covered by the tendons of abductor pollicis longus and extensor pollicis brevis. The lateral radiocarpal ligament is attached to its tip, and the palmar radiocarpal ligament is attached to the terminal ridge on the anterior surface of the lower end. The base of the triangular articular disc of the inferior radio-ulnar joint is attached to a smooth ridge distal to the ulnar notch, anterior to the lower end of the interosseous membrane. The lateral part of the carpal articular surface articulates with the scaphoid, and the medial part with the lateral part of the lunate. In full adduction, the proximal surface of the lunate is wholly in contact with the radius.

The radial dorsal tubercle receives a slip from the extensor retinaculum and is grooved medially by the tendon of extensor pollicis longus. The wide groove lateral to the tubercle contains the tendons of extensor carpi radialis longus laterally and extensor carpi radialis brevis medially, together with their synovial sheaths. Medially, the dorsal surface is grooved by the tendons of extensor digitorum, separated from the bone by the tendons of extensor indicis and the posterior interosseous nerve. The dorsal radiocarpal ligament is attached to the distal margin of this surface.

### Vascular supply

The radius is supplied by branches of the radial, ulnar, and anterior and posterior interosseous arteries through multiple metaphyseal nutrient foramina. Usually, there is one nutrient diaphyseal foramen located on the anterior surface of the bone, which is directed proximally towards the elbow. A network of small fascioperiosteal and musculoperiosteal vessels arise from the compartmental vessels and reach the bone via its septal and muscular attachments.

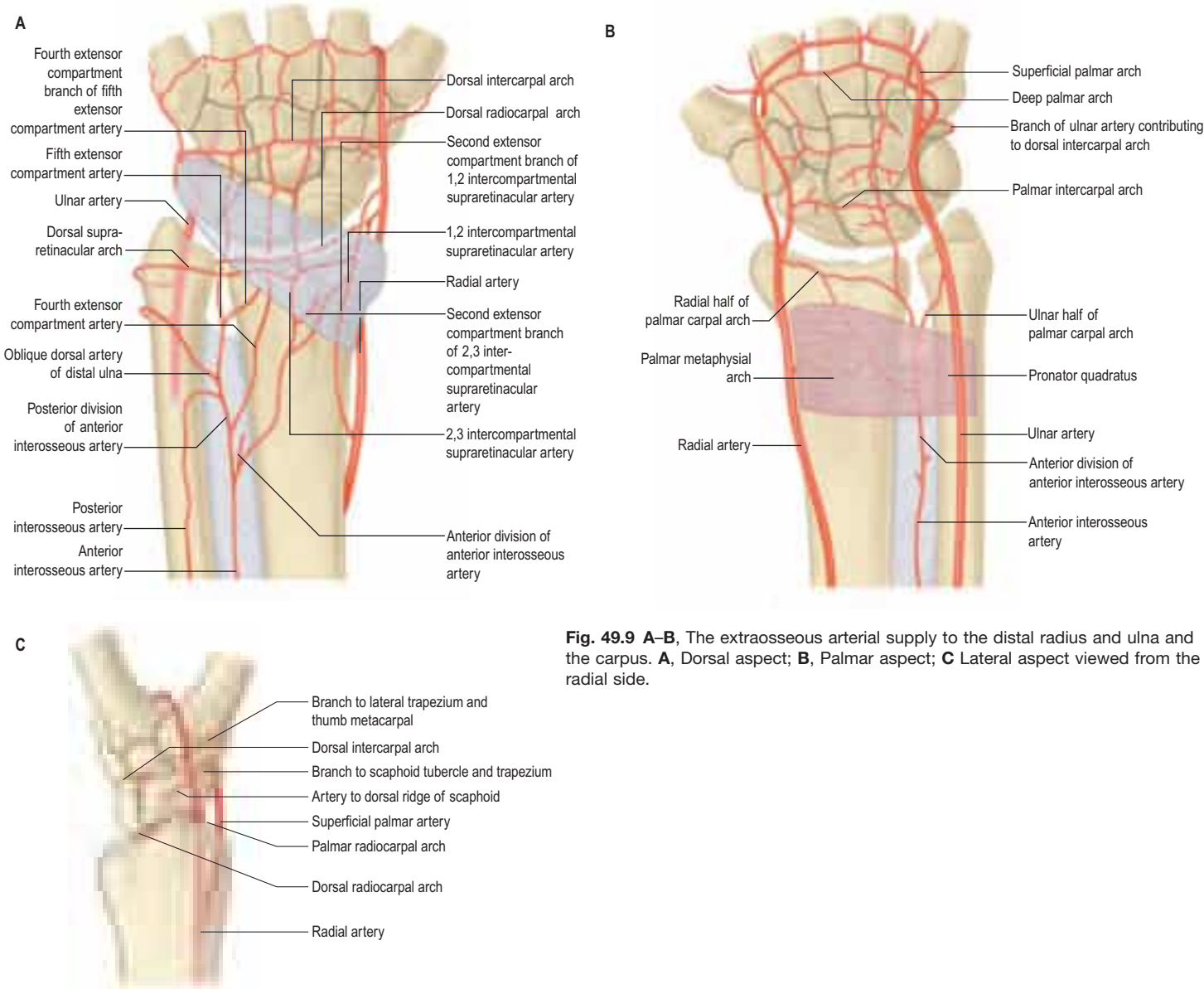
The dorsal metaphysis of the distal radius is richly supplied (Fig. 49.9A). Branches connect the anterior interosseous artery proximally to the dorsal carpal arch distally. These pass through the fourth and fifth extensor compartments of the wrist and provide metaphyseal nutrient arteries. Intercompartmental vessels send nutrient arteries to the radius through the retinaculum between the first and second dorsal compartments, and the second and third dorsal compartments. These vessels originate from the radial artery and anterior interosseous arteries, respectively, and anastomose with the dorsal carpal arch.

### Ossification

The radius ossifies from three centres. One appears centrally in the shaft in the eighth week of fetal life, and the others appear in each end (see Figs 48.13, 49.12). Ossification begins in the distal epiphysis towards the end of the first postnatal year and in the proximal epiphysis during the fourth year in females and the fifth year in males. The proximal



**Fig. 49.7** The proximal end of the left radius: anterior view. Key: 1, head; 2, radial tuberosity; 3, shaft; 4, neck.



**Fig. 49.9 A–B,** The extraosseous arterial supply to the distal radius and ulna and the carpus. **A,** Dorsal aspect; **B,** Palmar aspect; **C** Lateral aspect viewed from the radial side.

epiphysis fuses in the fourteenth year in females and seventeenth in males, and the distal in the seventeenth and nineteenth years, respectively. A fourth centre sometimes appears in the tuberosity at about the fourteenth or fifteenth year.

## ULNA

The ulna is medial to the radius in the supinated forearm. Its proximal end is a massive hook, concave forwards, that articulates with the distal humerus (Figs 49.10–49.11). The lateral border of the shaft is a sharp interosseous crest. The bone diminishes progressively from its proximal mass throughout almost its whole length, but, at its distal end, expands into a small, rounded head and styloid process. The shaft is triangular in cross-section proximally, and almost cylindrical distally. Along its whole length, the ulna is slightly convex posteriorly. Mediolaterally, its profile is sinuous; the proximal half has a slight curvature that is concave laterally and the distal half a curvature that is concave medially.

### Proximal ulna

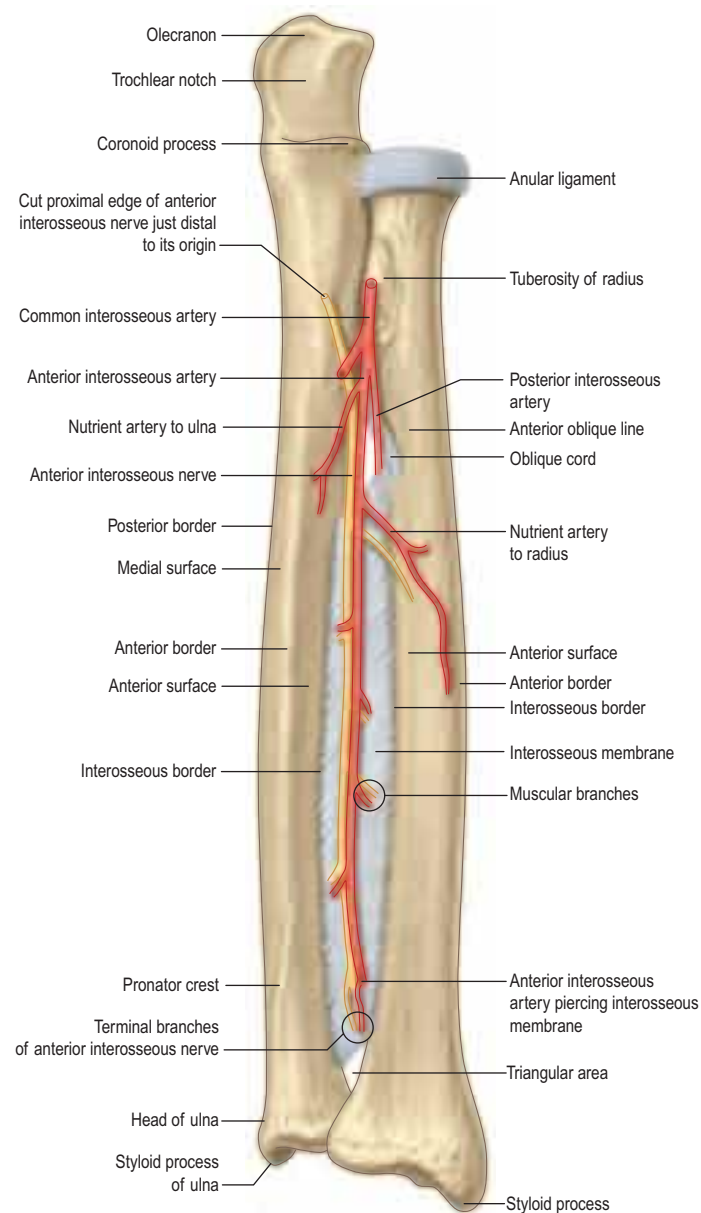
The proximal end of the ulna has large olecranon and coronoid processes, and trochlear and radial notches that articulate with the humerus and radius, respectively. The olecranon is more proximal and is bent forwards at its summit like a beak that enters the olecranon fossa of the humerus in extension. Its posterior surface is smooth, triangular and subcutaneous, and its proximal border underlies the ‘point’ of the elbow. In extension, it can be felt near a line joining the humeral epicondyles, but in flexion, it descends, so that the three osseous points form an isosceles triangle. Its anterior, articular surface forms the

proximal area of the trochlear notch. Its base is slightly constricted where it joins the shaft and is the narrowest part of the proximal ulna. The coronoid process projects anteriorly distal to the olecranon. Its proximal aspect forms the distal part of the trochlear notch. On the lateral surface, distal to the trochlear notch, a shallow, smooth, oval radial notch articulates with the radial head. Distal to the radial notch, the surface is hollow to accommodate the radial tuberosity during pronation and supination. The anterior surface of the coronoid is triangular. Its distal part is the tuberosity of the ulna and its medial border is sharp and bears a small tubercle proximally.

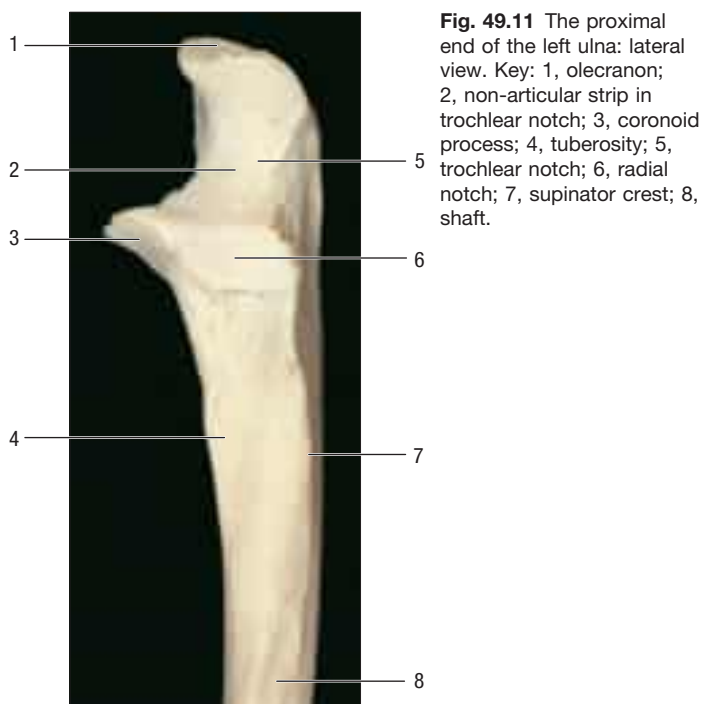
The trochlear notch articulates with the trochlea of the humerus. It is constricted at the junction of the olecranon and coronoid processes, where their articular surfaces may be separated by a narrow, rough, non-articular strip. A smooth ridge, adapted to the groove on the humeral trochlea, divides the notch into medial and lateral parts. The medial fits into the trochlear flange. The radial notch, an oval or oblong proximal depression on the lateral aspect of the coronoid process, articulates with the periphery of the radial head and is separated from the trochlear notch by a smooth ridge.

### Ulnar shaft

The ulnar shaft is triangular in cross-section in its proximal three-quarters, but distally is almost cylindrical (see Figs 49.31–49.32). It has anterior, posterior and medial surfaces and interosseous, posterior and anterior borders. The interosseous border is a conspicuous lateral crest in its middle half. Proximally, it becomes the supinator crest, which is continuous with the posterior border of a depression distal to the radial notch. Distally, it disappears. The rounded anterior border starts medial to the ulnar tuberosity, descends backwards, and is usually traceable to the base of the styloid process. The posterior border, also rounded,



**Fig. 49.10** The interosseous membrane and oblique cord of the forearm: anterior aspect. Membrane and cord are syndesmoses.



**Fig. 49.11** The proximal end of the left ulna: lateral view. Key: 1, olecranon; 2, non-articular strip in trochlear notch; 3, coronoid process; 4, tuberosity; 5, trochlear notch; 6, radial notch; 7, supinator crest; 8, shaft.

descends from the apex of the posterior aspect of the olecranon, and curves laterally to reach the styloid process. It is palpable throughout its length in a longitudinal furrow, most obvious when the elbow is fully flexed.

The anterior surface, between the interosseous and anterior borders, is longitudinally grooved, sometimes deeply (see Fig. 49.10). Proximal to its midpoint, a proximally directed nutrient foramen admits a branch of the anterior interosseous artery. Distally, it is crossed obliquely by a rough, variable prominence, descending from the interosseous to the anterior border. The medial surface, between the anterior and posterior borders, is transversely convex and smooth. The posterior surface, between the posterior and interosseous borders, is divided into three areas. The most proximal is limited by a sometimes faint, oblique line that ascends laterally from the junction of the middle and upper thirds of the posterior border to the posterior end of the radial notch. The region distal to this line is divided into a larger medial and narrower lateral strip by a vertical ridge, which is usually distinct only in its proximal three-quarters.

### Distal ulna

The distal end of the ulna is slightly expanded and has a head and styloid process. The head is palpable and visible in pronation on the posteromedial aspect of the wrist. Its lateral convex articular surface fits the radial ulnar notch. Its smooth distal surface is separated from the carpus by an articular disc, the apex of which is attached to a rough area between the articular surface and styloid process. The ulnar styloid process is a short, round, posterolateral projection of the distal end of the ulna, palpable in supination about 1 cm proximal to the plane of the radial styloid. A posterior vertical groove is present between the head and styloid process.

### Muscular, ligamentous and articular attachments

The capsule of the elbow joint is attached to the proximal, anterior surface of the olecranon, and the tendon of triceps is attached to its rough posterior two-thirds; the capsule and tendon may be separated by a bursa. The medial surface of the olecranon is marked proximally by the attachment of the posterior and oblique bands of the ulnar collateral ligament and by the ulnar part of flexor carpi ulnaris. The smooth area distal to this is the most proximal attachment of flexor digitorum profundus. Anconeus is attached to the lateral surface of the olecranon and the adjoining posterior surface of the ulnar shaft as far as its oblique line. The posterior surface of the ulna is separated from the skin by a subcutaneous bursa.

Brachialis is attached to the anterior surface of the coronoid process, including the ulnar tuberosity. The oblique and anterior bands of the ulnar collateral ligament and the distal part of the humero-ulnar slip of flexor digitorum superficialis are attached to a small tubercle at the proximal end of the medial border. Further distally, the margin of the coronoid process provides attachment for the ulnar head of pronator teres. An ulnar head of flexor pollicis longus may be attached to the lateral or, more rarely, the medial, border of the coronoid process. Fibres of flexor digitorum profundus are attached to its medial surface. The anular ligament is attached to the anterior rim of the radial notch and, posteriorly, to a ridge at, or just behind, the posterior margin of the notch. The depressed area distal to the notch is limited behind by the supinator crest, and both provide attachment for supinator.

The olecranon area of the trochlear notch is usually divided into three areas. The most medial faces anteromedially, and is grooved to fit the medial flange of the humeral trochlea, with which it makes increasing contact during flexion. A flat intermediate area fits the lateral flange. The most lateral area is a narrow strip that abuts the trochlea in extension. The articular surface is narrower than the base of the olecranon; non-articular areas are related to the synovial processes. The coronoid area of the trochlear notch is also divided; its medial and lateral areas correspond to the medial and intermediate areas of the olecranon. The medial is the deeper, and conforms to the convex medial trochlear flange. The medial and anterior parts of the capsular ligament are attached to its medial and anterior borders.

The deep fascia of the forearm is attached to the subcutaneous posterior border, which also provides an attachment for the aponeurosis of flexor digitorum profundus in its proximal three-quarters, for flexor carpi ulnaris in its proximal half, and for extensor carpi ulnaris in its middle third. These three muscles are attached to the posterior border through a common blended aponeurosis. The interosseous membrane is attached along the interosseous border, except for a short length proximally.

Flexor digitorum profundus is attached to the proximal three-quarters of the anterior border and medial surface, arising medial to the



coronoid process and olecranon. The rough strip across the distal fourth of the anterior surface provides part of the bony attachment for pronator quadratus. Anconeus is attached to the posterior surface proximal to the oblique line and lateral to the olecranon. The narrow strip between the interosseous border and vertical ridge gives rise to the attachment of three deep muscles: abductor pollicis longus arises from the proximal quarter, extensor pollicis longus from the succeeding quarter (sometimes a ridge is interposed between them), and extensor indicis from the third quarter. Extensor carpi ulnaris covers the broad strip medial to the vertical ridge and its tendon grooves the posterior aspect of the distal end of the ulna.

### Vascular supply

The ulna receives its vascular supply from branches of the radial, ulnar, anterior and posterior interosseous arteries via multiple metaphyseal nutrient foramina. Usually one, but occasionally two, major nutrient diaphyseal foramina are located on the anterior surface of the bone, directed proximally towards the elbow. A network of small fascioperiosteal and musculoperiosteal branches given off from the compartmental vessels reaches the bone via septal and muscular attachments.

### Ossification

The ulna ossifies from four main centres: one each in the shaft and distal end, and two in the olecranon (see Fig. 48.13; Fig. 49.12). Ossification begins in the midshaft about the eighth fetal week and extends rapidly. A centre appears in the distal end in the fifth year in females

and sixth year in males, and extends into the styloid process. The distal olecranon is ossified as an extension from the shaft, the remainder from two centres, one for the proximal trochlear surface and the other for a thin, scale-like proximal epiphysis on its summit. The proximal epiphysis appears in the ninth year in females and eleventh year in males, and the whole proximal epiphysis has usually joined the shaft by the fourteenth year in females and sixteenth year in males. The distal epiphysis usually unites with the shaft in the seventeenth year in females and eighteenth in males. Along with the distal humerus, the proximal ulna ossifies in a sequential manner; skeletal maturity may be assessed by observations of elbow radiographs.

## JOINTS

The elbow joint acts as a platform for the forearm and hand. The humero-ulnar and humero-radial joints form a complex hinge that enables flexion and extension of the forearm. The humero-radial joint, in conjunction with the proximal and distal radio-ulnar joints, allows full pronation and supination of the forearm. The configuration of the humero-ulnar joint, between the trochlea and the trochlear notch of the ulna and the distal prolongation of the medial edge of the trochlear pulley, tilts the ulna, forming a 'carrying angle' (approximately 14° in women and 11° in men) between the longitudinal axes of the humerus and ulna when the elbow is fully extended and supinated. These axes become aligned in full flexion.

## HUMERO-ULNAR JOINT

The humero-ulnar joint is a complex hinge joint and is continuous with two pivot joints between the distal humerus and proximal radius, and between the proximal radius and ulna (Figs 49.13, 49.15–17). There is a high degree of congruity between the deep trochlea of the humerus and the large trochlear notch of the ulna. This stable arrangement is enhanced by strong medial and lateral collateral ligaments and, in full extension, by the anterior capsule. An arc of 150° of flexion and extension is possible. A range of 180° of pronation and supination is achieved by synchronous movements at the proximal and distal radio-ulnar joints, aided by minor contributions from the radiocarpal and carpal joints.

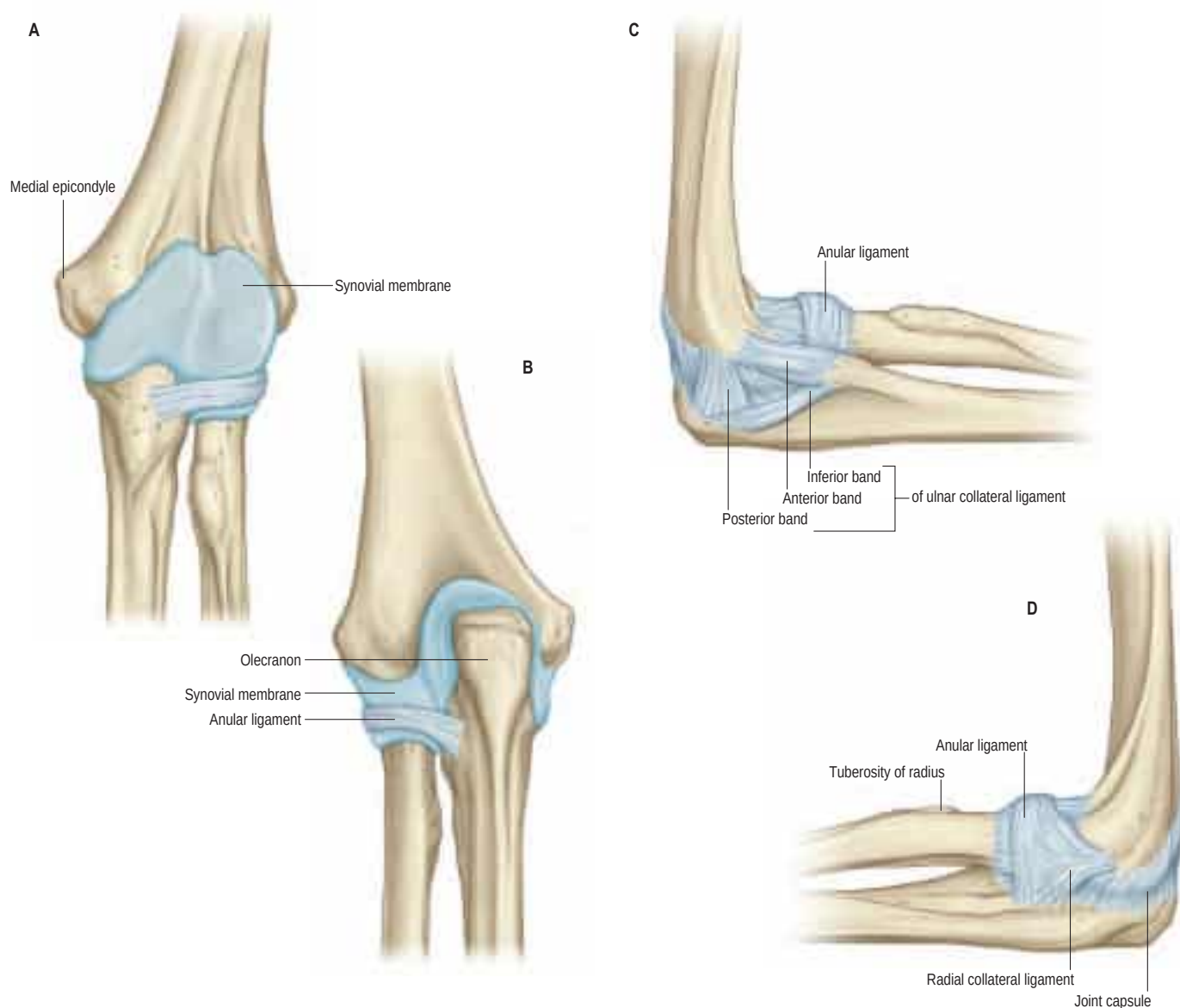
**Articulating surfaces** The distal humerus possesses two distinct articulating surfaces: the humeral trochlea and the capitulum (see Fig. 49.3). The shape of the humeral trochlea is asymmetric, with a longer medial flange. It is widest posteriorly in the adult. The joint line lies 2 cm distal to the interepicondylar line and is tilted inferomedially. The flexion–extension arc of elbow movement is not a simple hinge. Different parts of the distal humerus are in contact with the corresponding articular surface of the proximal ulna throughout the range of movement; in full extension of the elbow, a medial segment of the distal humerus is not in contact with the ulna, whereas in full flexion, a lateral segment is not in contact.

The surface of the trochlear notch has an asymmetrical sellar surface that allows some rotation during swing in the flexion–extension plane. It is predominantly concave in the transverse plane, but convex in the anteroposterior plane. There is a 'bare area' devoid of hyaline cartilage between the olecranon and coronoid areas of the trochlear notch that is covered by fibrofatty tissue and synovium. The humeral capitulum is reciprocally convex with the concavity of the articulating surface of the radial head. The more prominent medial rim of the radial head sits in the groove between the capitulum and trochlea. Contact between the capitulum and radial head is maximal with the elbow in mid-flexion and the forearm in mid-pronation.

**Fibrous capsule** The fibrous capsule is broad and thin anteriorly. It is attached proximally to the front of the medial epicondyle and humerus above the coronoid and radial fossa, and distally to the edge of the ulnar coronoid process and anular ligament. It may be augmented by fibres from brachialis and blends into the ulnar collateral ligament medially and the radial collateral ligament laterally. Posteriorly, the capsule is even thinner and is related to the tendon of triceps and anconeus. It is attached proximally to the superoposterior aspect of the humeral capitulum, the posterior aspect of the medial epicondyle and the olecranon fossa. Only the lowest part of the olecranon fossa is devoid of a capsular attachment. Inferomedially, it reaches the superior and lateral margins of the olecranon and is laterally continuous with the superior radio-ulnar capsule deep to the anular ligament. The olecranon bursa lies between the capsule of the elbow and the



**Fig. 49.12** Radiographs of the forearm of a girl aged 11 years. **A**, Anteroposterior view. Key: 1, cartilaginous growth plates; 2, conjoined epiphyses of capitulum and lateral epicondyle; 3, proximal radial epiphysis; 4, distal radial epiphysis; 5, epiphysis of medial epicondyle; 6, diaphyseal bone; 7, trochlear epiphysis; 8, distal ulnar epiphysis. **B**, Lateral view, semi-flexed at the elbow. Note the epiphyses and adjacent radiotranslucent growth cartilages. Key: 1, olecranon; 2, distal ulnar epiphysis; 3, proximal radial epiphysis; 4, distal radial epiphysis.



**Fig. 49.13 A–D**, The left elbow joint. **A**, The synovial cavity partially distended: anterior aspect. The fibrous capsule of the elbow joint has been removed but the thick part of the anular ligament has been left *in situ*. Note that the synovial membrane (blue) descends below the lower border of the anular ligament. **B**, The synovial cavity partially distended: posterior aspect of the specimen represented in **A**. **C**, Medial aspect. **D**, Lateral aspect.

triceps insertion. Inflammation caused by pressure, friction or infection of the bursa can result in marked swelling and limitation in elbow function.

**Ligaments** The humero-ulnar and humero-radial articulations have ulnar and radial collateral ligaments (see Fig. 49.13).

**Ulnar collateral ligament** This triangular ligament consists of thick anterior, posterior and inferior bands connected by thinner tissue. The strongest and stiffest anterior band is attached by its apex to the anterior aspect of the medial epicondyle and by its broader base to a proximal tubercle on the medial margin of the coronoid. The anterior band is taut throughout most of the range of flexion and contributes to the stability of the joint. The posterior band of the ulnar collateral ligament is also triangular in shape; its apex attaches to the back of the medial epicondyle and its distal attachment is the medial margin of the olecranon. The posterior band becomes taut between 90° and full flexion of the elbow. Connecting these two thick bands is a less sturdy inferior oblique band that runs between the olecranon and coronoid processes, converting a depression on the medial margin of the trochlear notch into a foramen, through which the intracapsular fat pad is continuous with the extracapsular fat medial to the joint. The ulnar collateral ligament of the elbow lies beneath triceps, flexor carpi ulnaris, the ulnar nerve and the attachment of flexor digitorum superficialis as it extends from the medial epicondyle to the medial coronoid border.

**Radial collateral ligament** The radial collateral ligament is attached low on the lateral epicondyle and to the anular ligament. Some of the posterior fibres cross the ligament to the proximal end of the supinator crest of the ulna. The ligament blends with the bony attachments of supinator and extensor carpi radialis brevis. It is taut throughout most of the range of flexion of the elbow and contributes to the stability of the joint.

**Synovial membrane** The synovial membrane extends from the articular margins of the humerus and lines the coronoid, radial and olecranon fossae, the flat medial surface of the trochlea, the deep surface of the elbow capsule and the distal part of the anular ligament. A crescentic synovial fold projecting between the proximal radius and ulna contains extrasynovial fat and forms the division between the humero-ulnar and humero-radial joints.

There are three other fat pads between the capsule and synovium; the largest is in the olecranon fossa, where it is pushed into the fossa by triceps during flexion of the elbow. The two other fat pads are in the coronoid and radial fossae, and are pushed into these fossae by brachialis on extension of the elbow. Smaller fat pads covered by synovium project into the joint at the trochlear notch and also cover non-articular areas of bone.

**Vascular supply** The articular arteries of the elbow are derived from numerous small periarticular anastomoses around the joint (see Fig. 49.29).



**Innervation** The elbow joint is innervated by filaments arising from all of the main nerves of the upper limb, except for the axillary nerve. The ulnar posterior part is supplied by the ulnar nerve and the medial cutaneous nerve of forearm; the ulnar anterior part is supplied by the median and musculocutaneous nerves. The radial nerve is the sole supplier of the radial posterior part, and the radial anterior part is innervated by both radial and musculocutaneous nerves (de Kesel et al 2012).

## PROXIMAL RADIO-ULNAR JOINT

The proximal radio-ulnar joint is a uniaxial pivot joint (Fig. 49.14).

**Articulating surfaces** Articulating surfaces include the articulation between the circumference of the radial head and the fibro-osseous ring of the ulnar radial notch and anular ligament.

**Fibrous capsule** The fibrous capsule of the proximal radio-ulnar joint is continuous with that of the elbow and is attached to the anular ligament.

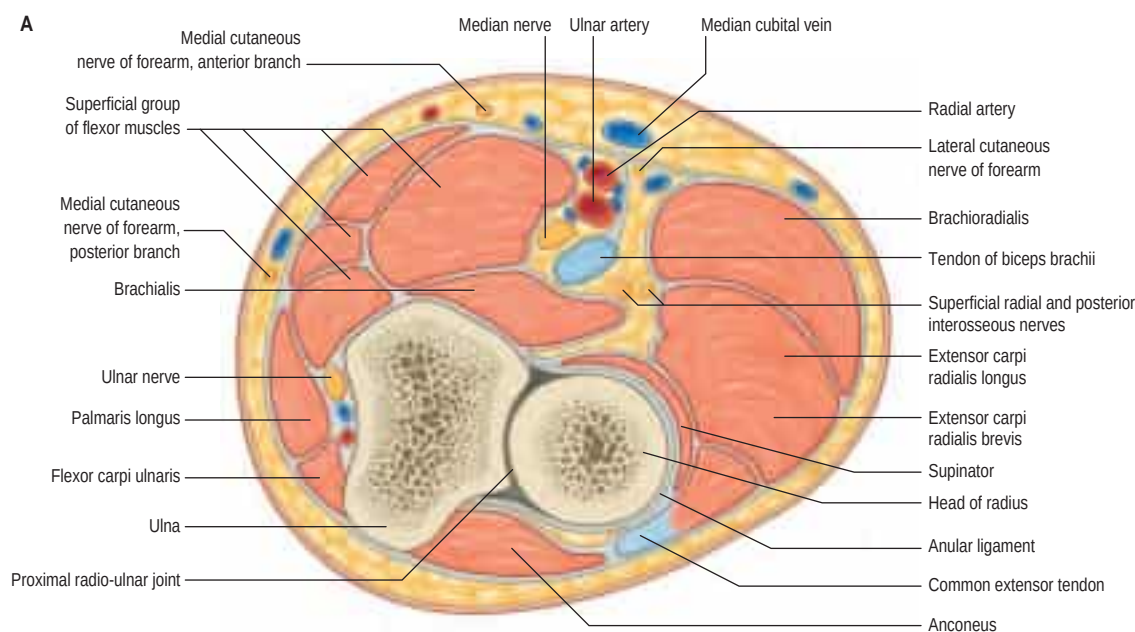
**Ligaments** The proximal radio-ulnar joint has anular and quadrate ligaments.

**Anular ligament** The anular ligament is the principal stabilizer of the joint. It is a strong composite band that winds around four-fifths of the

radial head and holds it against the radial notch of the ulna. It is attached to the anterior margin of the notch. The ligament is wider posteriorly, where it may be composed of several bands. It is attached to a rough ridge at or behind the posterior margin of the notch. Some bands may diverge and attach superiorly to the lateral margin of the trochlear notch, or inferiorly to the supinator crest. Proximally, the anular ligament blends into the joint capsule, except posteriorly where the capsule is deep to the ligament. Distally, the anular ligament gives off a few fibres, which pass over the synovial membrane to attach to the radial neck. The superficial surface of the anular ligament blends into the radial collateral ligament and provides an attachment for a part of supinator. The deep surface of the anular ligament is lined with a thin layer of cartilage where it is in contact with the radial head. Distally, there is a synovial membrane that is reflected up on to the radial neck. The anconeus and the interosseous recurrent artery lie behind the ligament.

**Quadrate ligament** The quadrate ligament is a much thinner and more fibrous structure than the anular ligament. It runs between the radial neck and the upper part of the supinator fossa of the ulna, and is taut throughout pronation and supination. It lies over the synovial membrane on the distal surface of the joint.

**Synovial membrane** The synovial membrane of the proximal radio-ulnar joint is continuous with that of the elbow joint, forming one continuous synovial cavity. It attaches to the joint margins, lining



**Fig. 49.14** The proximal left radio-ulnar joint. **A**, A transverse section. **B**, An axial magnetic resonance imaging (MRI) scan.



the capsule and anular ligament. The quadrate ligament prevents herniation of the synovial membrane between the free edges of the anular ligament.

**Vascular supply** The vascular supply of the proximal radio-ulnar joint is derived from a rich periarticular anastomotic plexus (Fig. 49.18; see Fig. 49.19).

**Innervation** The proximal radio-ulnar joint is innervated by small branches arising from the musculocutaneous, median, radial and ulnar nerves.

## RADIO-ULNAR SYNDESMOSES

The radial and ulnar shafts are connected by two syndesmoses, an oblique cord and an interosseous membrane.

### Oblique cord

The oblique cord is a small, inconstant, flat fascial band on the deep head of supinator (see Fig. 49.10). It extends from the lateral side of the ulnar tuberosity to the radius a little distal to its tuberosity. Its fibres are at right angles to those in the interosseous membrane. Its functional significance is unclear.

### Interosseous membrane

The interosseous membrane is a broad, thin, collagenous structure (see Fig. 49.10). Its fibres slant distomedially between the interosseous borders of the radius and ulna, and its distal part is attached to the posterior division of the radial shaft. Two or three posterior bands occasionally descend distolaterally across the other fibres. The membrane is deficient proximally, starting 2 or 3 cm distal to the radial tuberosity, and broadest at midlevel. An oval aperture near its distal margin conducts the anterior interosseous vessels to the back of the forearm. The posterior interosseous vessels pass through a gap between its proximal border and the oblique cord.

The membrane provides attachments for the deep forearm muscles. Anteriorly, in its proximal three-quarters, it is related laterally to flexor pollicis longus and medially to flexor digitorum profundus; between them, it is related to the anterior interosseous vessels and nerve. In its distal quarter, it is related to pronator quadratus. Its posterior relations are supinator, abductor pollicis longus, extensors pollicis brevis, longus and indicis, and, near the carpus, the anterior interosseous artery and posterior interosseous nerve.

The interosseous membrane consists of several parts, the most important of which is the central band originally described by Hotchkiss et al (1989). This probably acts in conjunction with the radial head to restrain the radius from proximal migration and stabilize the forearm. Previously described as being at its most taut in almost every position of rotation of the forearm, it now appears to maintain the same tension throughout pronation and supination. Noda et al (2009) have described four other subsidiary portions.

## DISTAL RADIO-ULNAR JOINT

The distal radio-ulnar joint is a uniaxial pivot joint.

**Articulating surfaces** The articulating surfaces are between the convex distal head of the ulna and the concave ulnar notch of the radius; they are connected by an articular disc.

**Fibrous capsule** The fibrous capsule is thicker anteriorly and posteriorly, and more lax proximally.

**Triangular fibrocartilage complex and distal radio-ulnar ligaments** The triangular fibrocartilage complex (TFCC) is a ligamentous and cartilaginous structure that suspends the distal radius and ulnar carpus from the distal ulna. It stabilizes the ulnocarpal and radio-ulnar joints, transmits and distributes load from the carpus to the ulna, and facilitates complex movements at the wrist. By definition, it is made up of the triangular fibrocartilage proper (the articular disc), meniscus homologue (the ulnocarpal meniscus), ulnar collateral ligament, dorsal and palmar radio-ulnar ligaments, floor of extensor carpi ulnaris subsheath, and the ulnolunate and ulnotriquetral ligaments. The triangular fibrocartilage proper is a biconcave body composed of chondroid

fibrocartilage that extends across the dome of the ulnar head, ranging in thickness between 2 and 5 mm. The meniscal homologue is a broad expansion of loose connective tissue that runs from the articular disc to the ulnar styloid and the surrounding joint capsule; it is not a major contributor to the structural integrity of the triangular fibrocartilage complex.

The triangular fibrocartilage complex is attached to the ulnar aspect of the lunate fossa of the radius. Medially, it is inserted into the head of the ulna and the base of the ulnar styloid. Here, it is joined by fibres from the ulnar part of the ulnar styloid (the ulnar collateral ligament); thus thickened, it attaches to the lunotriquetral interosseous ligament, triquetrum, hamate and base of the fifth metacarpal. A constant perforation just distal to the level of the ulnar styloid leads to the prestyloid recess. The dorsal and palmar (volar) aspects of the triangular fibrocartilage complex are thickened to form the dorsal and palmar radio-ulnar ligaments, which are attached to the dorsal and volar ulnar corners of the distal radius and to the fovea of the ulnar head at the base of the ulnar styloid (Palmer and Werner 1981, Pacek and Buterbaugh 2015).

The triangular fibrocartilage complex receives its blood supply from the ulnar artery via its palmar and dorsal radiocarpal branches, the dorsal branch of the anterior interosseous artery and the palmar branch of the anterior interosseous artery. Only the outer 15–20% of the disc is vascularized; the rest is avascular (Thiru-Pathi et al 1986, Bednar et al 1991) (Fig. 49.19).

The triangular fibrocartilage complex acts as a cushion for the ulnar carpus, preventing ulnocarpal abutment and overloading of the ulna in ulnar deviation. It is a major stabilizer of the distal radio-ulnar joint. Degenerative perforation of the triangular fibrocartilage complex becomes more common with age and is a constant finding after the age of 50 years (Mikić 1978).

**Synovial membrane** The synovial membrane lines the capsule and forms a sacciform recess in front of the distal part of the interosseous membrane (see Fig. 50.41).

**Vascular supply** The arterial supply of the distal radio-ulnar joint is derived from the palmar and dorsal branches of the anterior interosseous artery, reinforced by the posterior interosseous and ulnar arteries.

**Innervation** The distal radio-ulnar joint is innervated by branches of the anterior and posterior interosseous nerves.

## MOVEMENTS AT THE HUMERO-ULNAR, HUMERO-RADIAL AND PROXIMAL AND DISTAL RADIO-ULNAR JOINTS

The main movements that occur at the humero-ulnar, humero-radial and proximal and distal radio-ulnar joints are flexion, extension, supination and pronation.

### Flexion and extension

The elbow joint has relatively stable semi-congruent articular surfaces that are assisted by strong static and dynamic stabilizers, and which allow a flexion–extension arc of approximately 150°. When the elbow is extended and supinated, the forearm forms a ‘carrying angle’ with the humerus of approximately 17°; the absolute angle varies from individual to individual. It exists because the medial edge of the trochlear projects distally by 6 mm and because the superior articular surface of the coronoid is oblique. The carrying angle is not present in flexion when the humerus and ulna are in the same plane. With the arm adducted and the elbow flexed, the little finger meets the clavicle. If the humerus is rotated laterally, the little finger reaches the anterior shoulder. Pronation of the extended forearm masks the carrying angle by lining up the arm, forearm and hand.

### Accessory movements



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### Pronation and supination

Movements at the humero-radial, proximal and distal radio-ulnar joints pronate and supinate the hand. In pronation, the radius turns

The surface topography of the humero-ulnar joint permits some rotation of the ulna against the humerus. The radius acts as a mobile strut within the forearm, such that rotation of the distal radius around the ulna carries the radiocarpal joint, wrist and hand into pronation or supination. The medial side of the head of radius is longer than the lateral side; this eccentric morphology causes a cam effect during rotation of the proximal radio-ulnar joint. The radius gives longitudinal stability to the forearm by protecting the ulna from side-to-side forces.

The interosseous membrane contributes to longitudinal stability by virtue of its oblique course from the radius proximally to the ulna distally. It appears to maintain the same tension throughout pronation and supination (Ekenstam et al 1984). If the membrane is disrupted, as in an Essex-Lopresti injury (fracture of the head of the radius, disruption of the interosseous membrane and dislocation of the distal radio-ulnar joint), the radius will migrate proximally; persistent radio-ulnar dissociation causes a complex instability of the forearm (Hotchkiss et al 1989). Correction of the fixed deformity of the forearm commonly seen in untreated cases of spastic cerebral palsy or birth lesions of the brachial plexus usually requires release of the membrane because it has become relatively shortened. Pronator quadratus and supinator protect the interosseous membrane, to some extent, during forced, rapid rotation of the forearm by 'paying out' eccentrically, thereby reducing the load applied across the membrane.

The elbow is stabilized by the intrinsic shape of the joint and by its ligaments and muscles. The trochlear pulley is deeply seated within the trochlear notch, while the olecranon provides anteroposterior stability. Anterior stability in the sagittal plane is enhanced by the joint capsule, biceps, brachialis and the bicipital aponeurosis; posterior stability is enhanced by the triceps tendon and aponeurosis.

The radial and ulnar collateral ligaments are key to stability in the coronal plane. Resistance to varus strain comes from the radial ligament complex, of which the posterior band is the most important. Resistance to valgus strain is provided by the ulnar collateral ligament complex, of which the anterior band is the most important. If the head of the radius is lost, an intact ulnar collateral ligament maintains stability of the elbow. If the ligament is disrupted, the ulna drifts into valgus, there is rotational instability and the trochlea is damaged. Stability of the radius is enhanced by anconeus and by the muscles arising from the lateral aspect of the distal humerus. Muscles that arise from the medial epicondyle contribute to stability of the ulna. Muscles that act across the elbow have a shunting effect, creating articular joint reaction compression forces that contribute to close-packing (the position of a joint in which the greatest stability is achieved for the least expended effort). The humero-ulnar joint is most closely packed and, therefore, most resistant to fatigue in flexion and supination.

The flexion-extension arc of the elbow is accompanied by some rotation; the ulna is slightly pronated in extension and supinated in flexion. In full extension, the radial head can be palpated posteriorly because the radial facet is larger than the capitulum. The excursion of the radial head in extension is limited by tension in the posterior capsule, by the muscles anterior to the joint and by the tip of the olecranon entering the olecranon fossa. In flexion, the rim of the radial head enters the radial fossa and the tip of the coronoid process enters the coronoid fossa. Flexion is principally limited by the apposition of soft tissues.

Accessory movements include anterior and posterior translation of the radial head on the ulnar radial notch, and of the ulnar head on the radial ulnar notch.

Accessory movements of the human elbow are minimal and facilitate the movements described above. The ulna rotates slightly at the elbow and allows a few degrees of abduction and adduction. The radial head exhibits a little anteroposterior translation on the capitulum when it moves on the ulnar radial notch and the annular ligament moves back and forth; this occurs maximally with the elbow in 90° of flexion.



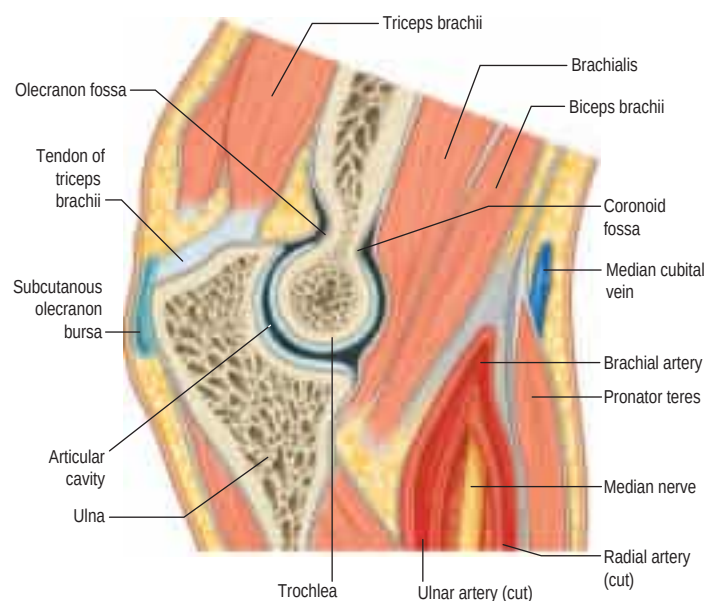


**Fig. 49.15** Radiographs of the left elbow joint of an adult. **A**, Anteroposterior view. Key: 1, olecranon fossa; 2, medial humeral epicondyle; 3, shadow of olecranon superimposed on trochlea; 4, humero-ulnar joint; 5, radial head articulating with radial notch of ulna; 6, lateral humeral epicondyle; 7, capitulum; 8, humero-radial joint; 9, head of radius; 10, radial tuberosity. **B**, Lateral view. Key: 1, head of radius; 2, profile of capitulum; 3, profile of trochlea; 4, olecranon.

anteromedially and obliquely across the ulna. Its proximal end remains lateral to the ulna while its distal end lies medially and the interosseous membrane becomes spiralled. In supination, the radius returns to its former position lateral and parallel to the ulna and the interosseous membrane becomes unspiralled. The hand can be turned through 140–150° with the elbow extended; this can be increased to nearly 360° by humeral rotation and movements of the scapula. Power is greater in supination, a fact that has affected the design of nuts, bolts and screws, which are tightened by supination in the right-handed.

### Stability of the distal radio-ulnar joint

With the wrist in neutral extension and deviation, the main axis for transmission of force from the hand to the forearm passes through the head of the capitate, then through the scapholunate ligament and finally into the distal radius. Loading is transferred to the triangular fibrocartilage complex and distal ulna in ulnar deviation. The joint is stabilized by extrinsic and intrinsic structures.



**Fig. 49.16** A sagittal section through the left elbow joint: medial aspect. The synovial membrane is shown in blue. (With permission from Waschke J, Paulsen F (eds), Sobotta Atlas of Human Anatomy, 15th ed, Elsevier, Urban & Fischer. Copyright 2013.)



**Fig. 49.17** A sagittal T1 weighted MRI section through the left elbow joint.

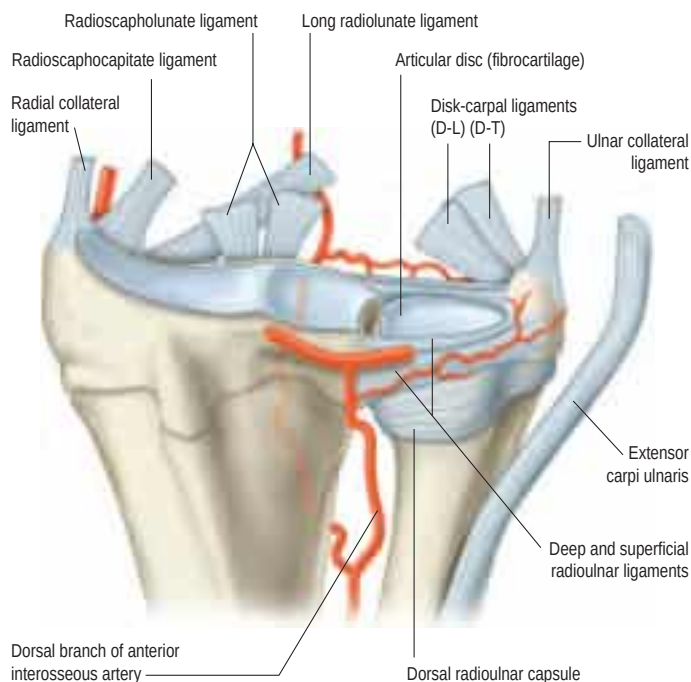
**Extrinsic stabilizers** Extrinsic stabilizers include the tendon of extensor carpi ulnaris, pronator quadratus, the interosseous membrane and the sheath of the sixth dorsal compartment, which overlies extensor carpi ulnaris (Spinner and Kaplan 1970).

**Intrinsic stabilizers** Intrinsic stabilizers are stronger than extrinsic stabilizers. They include the radio-ulnar capsule and the dorsal and palmar radio-ulnar ligaments. The dorsal and palmar radio-ulnar ligaments contain superficial and deep components. The superficial layers insert on to the ulnar styloid. The stronger deep layers insert more laterally into the fovea adjacent to the articular surface of the pole of the distal ulna (Kleinmann 2010).

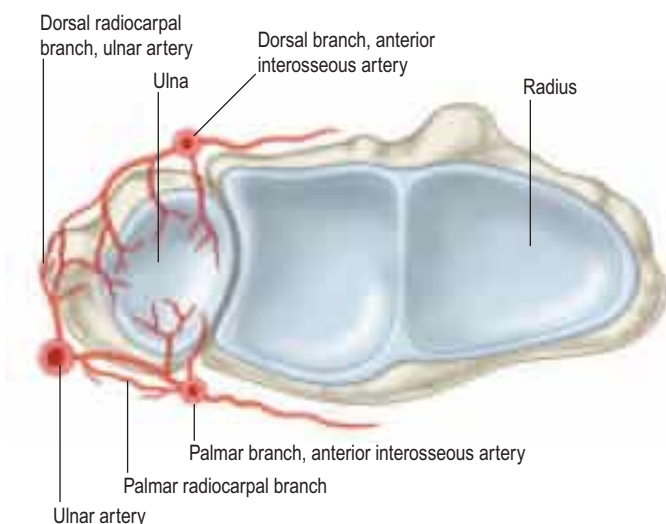
### Muscles producing movement

**Flexion** Movement is produced by brachialis, biceps brachii and brachioradialis. In slow flexion or its maintenance against gravity, brachialis and biceps are principally involved, even for light loads. With

Rotation of the forearm occurs between the head of the radius and the radial notch of the ulna proximally and the articulation of the head of the ulna and sigmoid notch. These proximal and distal radio-ulnar joints are pivot-type synovial joints; they act as a unit, permitting stable rotary motion (pronation  $61-66^\circ$ , supination  $70-77^\circ$ ). During rotation, the distal radius moves around the head of the ulnar. The axis for pronation and supination is often represented as a line through the centre of the radial head proximally and the ulnar attachment of the articular disc distally. This represents the axis of movement of the radius relative to the ulna but this does not remain stationary. The radial head rotates in the fibro-osseous ring; its distal lower end and articular disc swing round the head of the ulna. Rotation of the radial head occurs on the humeral capitulum. As the forearm moves from full pronation into supination, the ulna translocates medially by 9–10 mm, so that the axis of rotation shifts but still passes through the head of the ulna. In addition, the sigmoid notch changes its position of contact with the ulnar head, lying dorsal proximally in pronation and volar distally in supination. The distal end of the ulna is not stationary during these movements; it moves along a curved course, posterolaterally in pronation and anteromedially in supination. The axis of movement, as defined above, is therefore displaced laterally in pronation and medially in supination. Hence, the axis for pronation and supination of the forearm and hand passes between the bones at both the proximal and distal radio-ulnar joints when ulnar movement is marked, but through the centres of the radial head and ulnar styloid when it is minimal. The axis may be extended along any digit, depending on whether the distal end of the ulna is medially or laterally displaced. The hand will rotate further than the forearm because of the sliding-rotatory movement that occurs between the carpal bones and the bases of the metacarpals, and, to a very minor degree, at the radiocarpal joint.



**Fig. 49.18** The prime intrinsic stabilizer of the distal radio-ulnar joint is the triangular fibrocartilage. The triangular fibrocartilage complex consists of superficial and deep radio-ulnar fibres, the two disc carpal ligaments (disc-lunate (D-L) and disc-triquetral (D-T)), and the central articular disc. The articular disc is responsible for transferring load from the medial carpus to the pole of the distal ulna. The vascularized, peripheral radio-ulnar ligaments are nourished by dorsal and palmar branches of the posterior interosseous artery and are responsible for guiding the radiocarpal unit around the seat of the ulna. (Redrawn with permission from Kleinmann WB 2010 Stability of the distal radioulnar joint: biomechanics, pathophysiology, physical diagnosis and restoration of function. In: Slutsky DJ (ed) Principle and Practice of Wrist Surgery. Chapter 5, pp. 41–45.)



**Fig. 49.19** The arterial anatomy of the triangular fibrocartilage complex. (Redrawn with permission from Thiru-Pathi RG, Ferlic DC, Clayton MI, McClure DC 1986 Arterial anatomy of the triangular fibrocartilage of the wrist and its surgical significance. Journal of Hand Surgery 11A, 258–263.)

increasing speed and activity, brachioradialis is increasingly prominent; its attachments determine that it acts most effectively in mid-pronation. Against resistance, pronator teres, longus and brevis, flexors carpi radialis and ulnaris, and extensor carpi radialis may contribute.

**Extension** Movement is produced by triceps, anconeus and gravity.

**Pronation** Pronator quadratus is aided in rapid movement and against resistance by pronator teres. Gravity also assists.

**Supination** Movement is produced by supinator, in slow unresisted movement and extension, assisted by biceps in fast movements in flexion, especially when resisted.

## MUSCLES OF THE FOREARM

There is some anatomical segregation by function of muscles that cross the elbow. Most of those that arise medially pass to the flexor aspect of the wrist, or stabilize or move the ulnar column of the carpus and the ulnar side of the hand. Those that arise laterally pass to the lateral column of the forearm and carpus, and to the extensor aspect of the wrist. The muscles that act on the thumb have a wide origin; the extensor and abductor muscles arise from the entire width of the interosseous membrane. Rotator muscles, such as supinator, pronator teres and pronator quadratus, pass between the radius and ulna, crossing the longitudinal axis of the forearm.

## ANTERIOR COMPARTMENT

The anterior compartment contains the flexor muscles of the forearm. These are arranged in superficial and deep groups.

### Superficial flexor (volar) compartment

Muscles of the superficial flexor compartment are pronator teres, flexor carpi radialis, palmaris longus, flexor digitorum superficialis and flexor carpi ulnaris.

#### Pronator teres

**Attachments** Pronator teres arises from two heads. The humeral origin of pronator teres is the larger and more superficial. It arises at the medial supracondylar ridge just proximal to the medial epicondyle from the common tendon of origin of the flexor muscles, from the intermuscular septum between it and flexor carpi radialis, and from the antebrachial fascia (see Fig. 49.6, Figs 49.20–49.22). The smaller ulnar head arises from the medial side of the coronoid process of the ulna, distal to the attachment of flexor digitorum superficialis, and joins the humeral head at an acute angle. This ulnar coronoid attachment may be absent in some individuals. The muscle passes obliquely across the forearm to end in a flat tendon that is attached to a rough area midway along the lateral surface of the radial shaft at the apex of its lateral curve (see Fig. 49.25). Accessory slips may arise from a supracondylar process of the humerus, if present, or from biceps, brachialis or the medial intermuscular septum.

**Relations** The median nerve usually enters the forearm between the two heads of pronator teres, separated from the ulnar artery by the ulnar head. The lateral border of pronator teres is the medial margin of the cubital fossa (see Fig. 46.29). Proximally, the medial border of pronator teres is related to the lateral aspect of flexor carpi radialis. Pronator teres is anterior to flexor digitorum superficialis.

**Vascular supply** The humeral head of pronator teres is supplied by the inferior ulnar collateral and anterior ulnar recurrent arteries. The ulnar head is supplied by the common interosseous artery, the mid portion of the muscle belly by direct branches from the ulnar artery, and the radial insertion by the radial artery.

**Innervation** Pronator teres is innervated by the median nerve, C6 and 7.

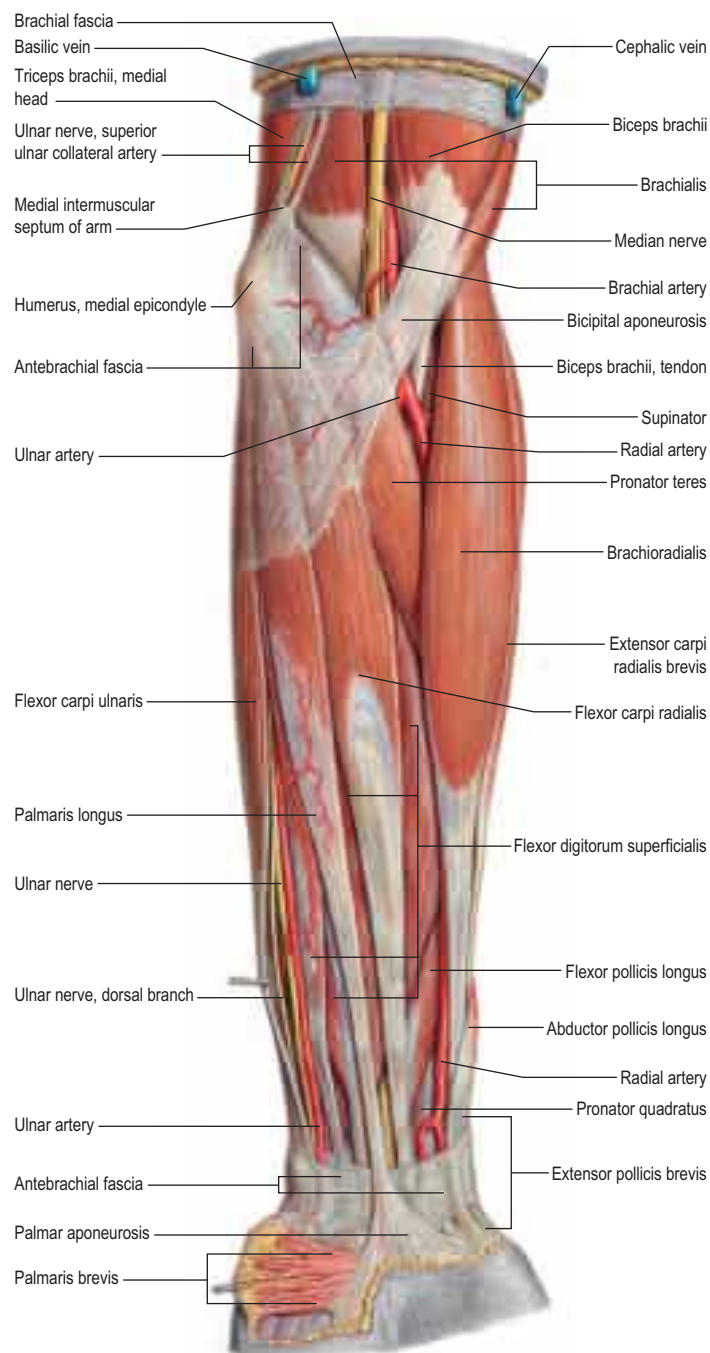
**Actions** Pronator teres pronates the forearm by rotating the radius medially on the ulna. It is a weak flexor of the elbow joint.

**Testing** Pronator teres is tested by palpating its contracting fibres during pronation against resistance.

#### Flexor carpi radialis

**Attachments** Flexor carpi radialis arises from the medial epicondyle via the common composite flexor tendon, from the antebrachial fascia and from adjacent intermuscular septa (see Fig. 48.11A). Its fusiform muscular belly converges in the mid-forearm into a long tendon that passes within a synovial sheath through a lateral canal, formed by the flexor retinaculum above and a groove on the trapezium beneath. It inserts on the palmar surface of the base of the second metacarpal and sends a slip to the third metacarpal. These distal attachments lie beneath the oblique head of adductor pollicis. Rarely, the muscle may be absent.





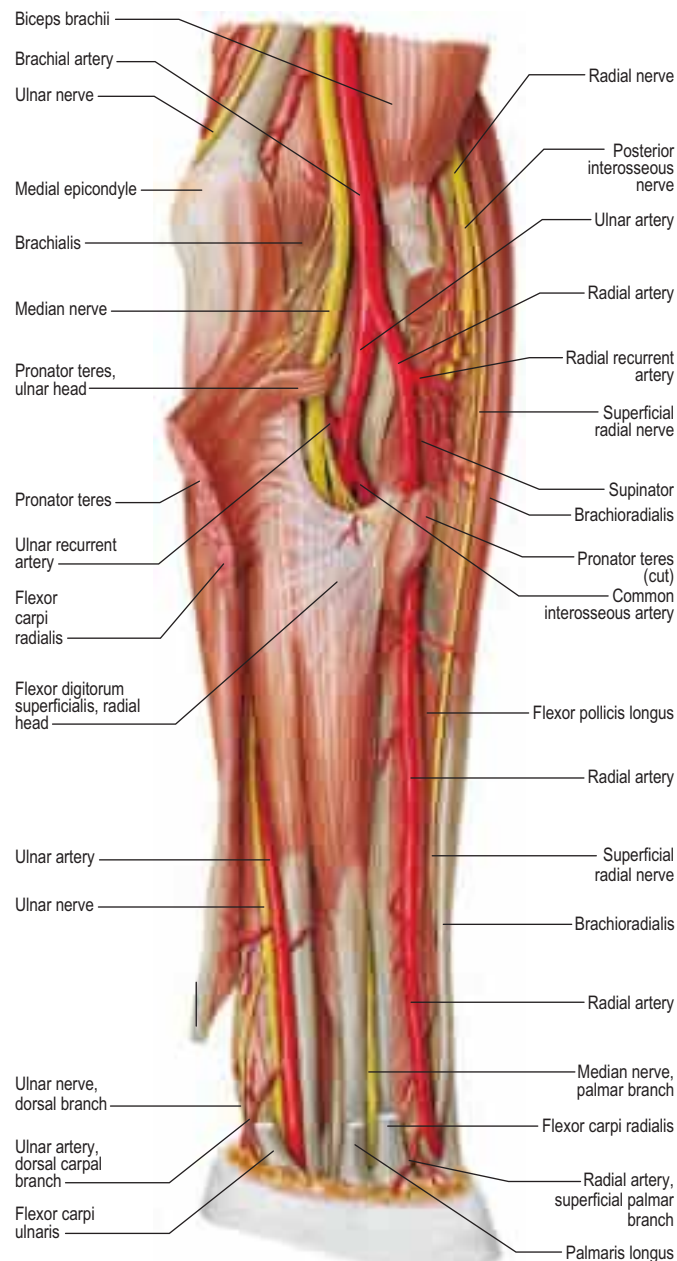
**Fig. 49.20** The superficial flexor muscles of the left forearm. (With permission from Waschke J, Paulsen F (eds), Sobotta Atlas of Human Anatomy, 15th ed, Elsevier, Urban & Fischer. Copyright 2013.)

There may be accessory slips from the biceps tendon, bicipital aponeurosis, coronoid process or radius. Distally, it may also be attached to the flexor retinaculum, trapezium or fourth metacarpal.

**Relations** At its origin, flexor carpi radialis lies medial to pronator teres. In the lower part of the forearm, the radial artery lies between the tendon of flexor carpi radialis and the tendon of brachioradialis; the radial pulse may be palpated proximal to the wrist between these two tendons.

**Vascular supply** Flexor carpi radialis is supplied by a single dominant proximal pedicle and several distal minor pedicles. The dominant pedicle is formed by a branch that arises from either the anterior or the posterior ulnar recurrent artery. The posterior ulnar recurrent artery passes deep to pronator teres to enter the deep surface of flexor carpi radialis; from here, it divides into a small ascending branch and a larger descending branch. The distal minor pedicles supplying flexor carpi radialis are six to eight branches from the radial artery that enter the muscle on the anterolateral side.

**Innervation** Flexor carpi radialis is innervated by the median nerve, C6 and 7.



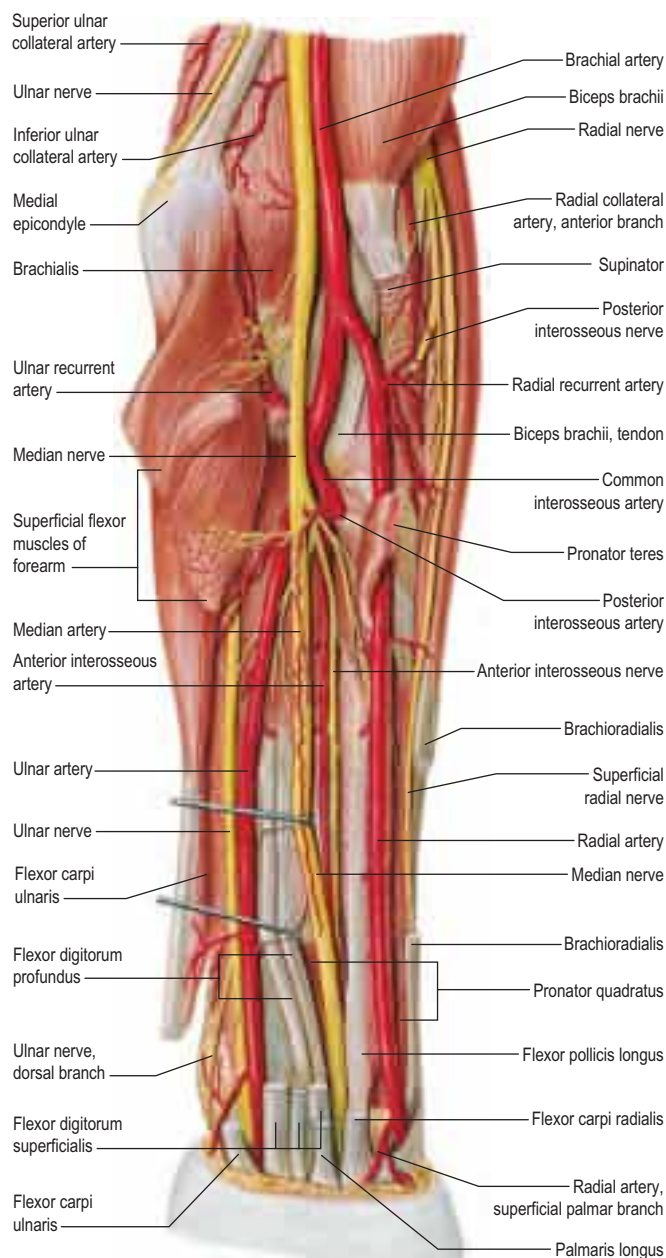
**Fig. 49.21** The deep flexor muscles of the left forearm. (With permission from Waschke J, Paulsen F (eds), Sobotta Atlas of Human Anatomy, 15th ed, Elsevier, Urban & Fischer. Copyright 2013.)

**Actions** Flexor carpi radialis is a wrist flexor. It acts with flexor carpi ulnaris to effect balanced wrist flexion, and in concert with the radial extensors of the wrist to achieve balanced abduction of the hand.

**Testing** Flexor carpi radialis is tested by palpating its contracting fibres during flexion of the wrist against resistance.

### **Flexor digitorum superficialis**

**Attachments** Flexor digitorum superficialis is the largest of the superficial flexors, and arises by two heads (see Figs 48.11A, 49.20–49.21). The humero-ulnar head arises from the medial epicondyle of the humerus via the common flexor origin, the anterior band of the ulnar collateral ligament, adjacent intermuscular septa, and from the medial side of the coronoid process proximal to the ulnar origin of pronator teres. The radial head, a thin sheet of muscle, arises from the anterior radial border between the radial tuberosity and the insertion of pronator teres. The median nerve and ulnar artery descend between the heads. The muscle usually separates into two strata, directed to digits two to five. The superficial stratum, joined laterally by the radial head, divides into two tendons for the middle and ring fingers. The deep stratum gives off a muscular slip to join the superficial fibres directed to the ring finger, and ends in two tendons for the index and little fingers. As the tendons pass behind the flexor retinaculum, they are arranged in pairs; the superficial pair passes to the middle and ring fingers, the deep pair to the index and little fingers (see Fig. 50.41). The four tendons diverge



**Fig. 49.22** The deep flexor aspect of the left forearm. (With permission from Waschke J, Paulsen F (eds), Sobotta Atlas of Human Anatomy, 15th ed, Elsevier, Urban & Fischer. Copyright 2013.)

distal to the carpal tunnel. Each passes towards a finger superficial to the corresponding tendon of flexor digitorum profundus. The two tendons for each finger enter the digital flexor sheath at the metacarpophalangeal joint. The superficialis tendon splits into two bundles that pass around the profundus to lie posteriorly, and then reunite to insert into the anterior surface of the middle phalanx. Some fibres interchange from one bundle to another.

An intermediate tendon is always found in the central branching area of the muscle belly and is an important landmark in deep dissection; it may be confused with the median nerve. The radial head of flexor digitorum superficialis may be absent and the muscular slip from the deep stratum may provide most or all of the fibres acting on the index finger. The fibres associated with the little finger may be absent, in which case they are replaced by a separate slip from the ulna, flexor retinaculum or palmar fascia. Variations occur in the arrangement of the tendons.

**Relations** Flexor digitorum superficialis runs deep to pronator teres and flexor carpi radialis. The median nerve and ulnar artery descend between the heads of flexor digitorum superficialis, and the nerve continues distally deep to the muscle.

**Vascular supply** The humeral head of flexor digitorum superficialis is supplied by the anterior ulnar recurrent artery. The main body of the muscle is supplied on its anterior surface by three or four branches from

both the ulnar and the radial arteries. The posterior surface is supplied by the ulnar artery and median artery, and the lateral surface by additional branches from the radial artery.

**Innervation** Flexor digitorum superficialis is innervated by the median nerve, C8 and T1.

**Actions** Flexor digitorum superficialis is a flexor of all the joints over which it passes, i.e. the wrist, metacarpophalangeal and proximal interphalangeal joints; its precise action depends on which other muscles are acting. It has independent muscle slips to all four fingers, and is therefore able to flex the proximal interphalangeal joints individually, unlike flexor digitorum profundus, which has a muscle group common to the middle, ring and little fingers.

**Testing** The independent action of flexor digitorum superficialis for a particular finger is tested by flexing that digit while holding the three other fingers in full extension. This eliminates any simultaneous contraction of the flexor digitorum profundus that might flex the digit.

### Palmaris longus

**Attachments** Palmaris longus arises from the medial epicondyle of the humerus via the common flexor origin and from adjacent intermuscular septa and deep fascia. It is a slender, fusiform muscle that runs medial to flexor carpi radialis (see Figs 48.11A, 49.20, 49.32). It becomes tendinous in the mid-forearm, and the long tendon passes anterior (superficial) to the flexor retinaculum. A few fibres leave the tendon and interweave with the transverse fibres of the retinaculum but most of the tendon passes distally. As the tendon crosses the retinaculum, it broadens out to become a flat sheet that becomes incorporated into the palmar aponeurosis. Palmaris longus is occasionally absent on one or both sides.

**Relations** Palmaris longus runs distally, flanked by flexor carpi radialis and flexor carpi ulnaris, and superficial to flexor digitorum superficialis. The median nerve at the wrist lies partly under the cover of the tendon of palmaris longus and partly between the tendons of palmaris longus and flexor carpi radialis.

**Vascular supply** The muscle belly of palmaris longus is supplied by a small branch from the anterior ulnar recurrent artery. A contribution is sometimes made by the median artery, if this is well developed.

**Innervation** Palmaris longus is innervated by the median nerve, C7 and 8.

**Actions** Palmaris longus is a weak wrist flexor. Through its insertion in the palmar aponeurosis, it also acts as an anchor for the skin and fascia of the hand against horizontal shearing forces.

**Testing** If the wrist is flexed against resistance, the taut tendon of palmaris longus will be seen in the midline of the flexor wrist crease as the tendon passes superficial to the flexor retinaculum. When testing palmaris longus, it helps to oppose the thumb to the middle fingertip at the same time as the wrist is flexed, in order to make palmaris stand out.

### Flexor carpi ulnaris

**Attachments** Flexor carpi ulnaris is the most medial of the superficial forearm flexors (see Figs 49.20, 49.31, 49.32). It arises by two heads, humeral and ulnar, connected by a tendinous arch. The smaller humeral head arises from the medial epicondyle via the common composite tendon (see Fig. 48.11A). The ulnar head has an extensive origin from the medial margin of the olecranon and proximal two-thirds of the posterior border of the ulna (from an aponeurosis shared with extensor carpi ulnaris and flexor digitorum profundus) and from the intermuscular septum between it and flexor digitorum superficialis (see Fig. 49.6). Occasionally, there is a slip from the coronoid process. A thick tendon forms along its anterolateral border in its distal half. The tendon is attached to the pisiform, and from here it is prolonged to the hamate and fifth metacarpal by pisohamate and pisometacarpal ligaments. The attachment to the flexor retinaculum and the fourth or fifth metacarpal bones is sometimes substantial.

**Relations** Flexor carpi ulnaris is the most medial of the forearm flexors and forms the bulk of the soft tissue cover of the medial aspect of the ulna. The ulnar nerve and posterior ulnar recurrent artery pass under the tendinous arch between its humeral and ulnar heads. Ulnar vessels and nerve lie lateral to its tendon in the distal forearm.



**Vascular supply** The main arterial supply of flexor carpi ulnaris is derived from three pedicles. The proximal pedicle arises from a branch of the posterior ulnar recurrent artery as it passes between the humeral and ulnar heads. The middle pedicle arises from the ulnar artery and enters the muscle at the junction of the upper and middle thirds. The distal pedicles arise from the ulnar artery and enter the muscle at the musculotendinous junction. Flexor carpi ulnaris also receives a small supply near its origin from the inferior ulnar collateral artery.

**Innervation** Flexor carpi ulnaris is innervated by the ulnar nerve, C7, 8 and T1.

**Actions** Flexor carpi ulnaris is a powerful flexor and also adductor (ulnar deviation) of the wrist. When it acts with flexor carpi radialis, balanced wrist flexion is achieved. When it acts with extensor carpi ulnaris, balanced adduction occurs.

**Testing** Flexor carpi ulnaris is tested by palpating its tendon while the wrist is flexed against resistance, or by palpating the tendon while the patient abducts the little finger against resistance. The muscle synergistically stabilizes the pisiform, giving abductor digiti minimi a stable origin.

## Deep flexor (volar) compartment

The muscles of the deep flexor compartment are flexor digitorum profundus, flexor pollicis longus and pronator quadratus (see Fig. 49.22).

### Flexor digitorum profundus

**Attachments** Flexor digitorum profundus arises deep to the superficial flexors from the proximal three-quarters of the anterior and medial surfaces of the ulna (see Figs 49.5, 49.6, 49.22). It embraces the attachment of brachialis above and extends distally almost to pronator quadratus. It also arises from a depression on the medial side of the coronoid process, from the proximal three-quarters of the posterior ulnar border by an aponeurosis that it shares with flexor and extensor carpi ulnaris, and from the anterior surface of the ulnar half of the interosseous membrane. The muscle ends in four tendons that run deep to the tendons of flexor digitorum superficialis and the flexor retinaculum (see Fig. 50.41). The part of the muscle that acts on the index finger is usually distinct throughout, while the tendons for the other fingers are interconnected by areolar tissue and tendinous slips as far as the palm. Anterior to their proximal phalanges, the tendons pass through the tendons of flexor digitorum superficialis to insert on the palmar surfaces of the bases of the distal phalanges. The tendons of the profundus undergo fascicular rearrangement as they pass through those of superficialis. Flexor digitorum profundus may be joined by accessory slips from the radius (which act on the index finger), flexor superficialis, flexor pollicis longus, the medial epicondyle or the coronoid process.

**Relations** Flexor digitorum profundus forms most of the surface elevation medial to the palpable posterior ulnar border. In the forearm, the median nerve runs on the anterior surface of its lateral aspect. In the palm, the lumbricals are attached to its tendons.

**Vascular supply** The origin of flexor digitorum profundus is supplied by the inferior ulnar collateral and ulnar recurrent arteries. The proximal part is supplied by one or two branches from either the ulnar or the common interosseous artery. The distal part is supplied by a series of branches from the ulnar artery, the anterior interosseous artery and the median artery.

**Innervation** The medial part of flexor digitorum profundus, i.e. the muscle bellies to the little and ring fingers, is innervated by the ulnar nerve. The lateral part, i.e. the muscle bellies to the middle and index fingers, is innervated by the anterior interosseous branch of the median nerve, C8 and T1.

**Actions** Flexor digitorum profundus flexes the distal interphalangeal joints of the fingers. It is also capable of flexing any or all of the joints over which it passes and therefore has a role in coordinated finger flexion. The index finger tendon is usually capable of independent function, whereas the other three work together.

**Testing** Flexor digitorum profundus is tested by flexing the distal interphalangeal joint while holding the proximal interphalangeal joint in extension.

### Flexor pollicis longus

**Attachments** Flexor pollicis longus arises from the grooved anterior surface of the radius, and extends from below its tuberosity to the upper attachment of pronator quadratus (see Figs 49.5, 49.21–49.22). It also arises from the adjacent interosseous membrane and, frequently, by a variable slip from the lateral, or more rarely medial, border of the coronoid process, or from the medial epicondyle of the humerus. The muscle ends in a flattened tendon that passes behind the flexor retinaculum, between opponens pollicis and the oblique head of adductor pollicis, to enter a synovial sheath. It inserts on the palmar surface of the base of the distal phalanx of the thumb.

Flexor pollicis longus can sometimes be connected to flexor digitorum superficialis or profundus, or pronator teres. The interosseous attachment, and, indeed, the whole muscle, may be absent. Anomalous tendon slips from the flexor pollicis longus to flexor digitorum profundus are common.

**Relations** Flexor pollicis longus runs lateral to flexor digitorum profundus. The anterior interosseous nerve and vessels descend on the interosseous membrane between these muscles.

**Vascular supply** The medial half of flexor pollicis longus is supplied by the anterior interosseous artery. The lateral half is supplied by branches of the radial artery. The median artery may contribute if it is well developed.

**Innervation** Flexor pollicis longus is innervated by the anterior interosseous branch of the median nerve, C7 and 8.

**Actions** Flexor pollicis longus principally flexes the phalanges of the thumb. It also flexes the carpometacarpal joint of the thumb, especially if the more distal joints are stiff or fused.

**Testing** Flexor pollicis longus is tested by flexing the interphalangeal joint of the thumb against resistance.

### Pronator quadratus

**Attachments** Pronator quadratus is a flat, quadrilateral muscle that extends across the front of the distal parts of the radius and ulna (see Figs 49.5, 49.27). It arises from the oblique ridge on the anterior surface of the shaft of the ulna, the medial part of this surface and a strong aponeurosis that covers the medial third of the muscle. The fibres pass laterally and distally to the distal quarter of the anterior border and surface of the shaft of the radius (see Fig. 50.30). Deeper fibres insert into the triangular area above the ulnar notch of the radius.

**Vascular supply** Pronator quadratus receives its main arterial supply from the anterior interosseous artery as it passes through the interosseous membrane.

**Innervation** Pronator quadratus is innervated by the median nerve, C7 and 8.

**Actions** Pronator quadratus is a pronator of the forearm. The deeper fibres oppose separation of the distal ends of the radius and ulna when axial load is transmitted through the carpus.

**Testing** Pronator quadratus is tested by pronation of the forearm against resistance while the wrist finger flexors are relaxed. The simultaneous contraction of pronator teres makes it difficult to test the independent action of pronator quadratus.

## POSTERIOR COMPARTMENT

The posterior compartment contains the extensor muscles of the forearm, brachioradialis and supinator. The superficial posterior muscles include anconeus, brachioradialis, extensor carpi radialis longus, extensor carpi radialis brevis, extensor digitorum, extensor digiti minimi and extensor carpi ulnaris. The deep posterior group of muscles includes supinator, abductor pollicis longus, extensor pollicis brevis, extensor pollicis longus and extensor indicis.

## Superficial extensor compartment

### Brachioradialis

**Attachments** Brachioradialis is the most superficial muscle along the radial side of the forearm and forms the lateral border of the cubital

fossa (see Figs 49.22, 49.25). It arises from the proximal two-thirds of the lateral supracondylar ridge of the humerus and the anterior surface of the lateral intermuscular septum (see Fig. 48.11A, 49.5). The muscle fibres end above mid-forearm level in a flat tendon that is inserted on the lateral side of the distal end of the radius, just proximal to the styloid process. This attachment may be more proximal; the muscle may fuse proximally with brachialis and the tendon may, occasionally, divide into two or three separately attached slips. In rare instances, it is double or absent.

**Relations** The radial nerve and the anastomosis between the profunda brachii artery and the radial recurrent artery lie in the groove between brachioradialis and brachialis, anterior to the lateral intermuscular septum. The tendon is crossed near its distal end by the tendons of abductor pollicis longus and extensor pollicis brevis. The radial artery is on its ulnar (medial) side. The superficial radial nerve emerges from between brachioradialis and extensor carpi radialis brevis to run superficially over the tendon of extensor carpi radialis brevis and extensor pollicis brevis, and then superficial to the extensor retinaculum.

**Vascular supply** Brachioradialis is supplied by branches of the radial recurrent artery that pierce the posteromedial surface of the muscle. It also receives branches from the radial collateral branch of the profunda brachii and directly from the radial artery in the distal part of the muscle.

**Innervation** Brachioradialis is innervated by the radial nerve, C5 and 6.

**Actions** Brachioradialis is a powerful flexor of the elbow, acting most strongly with the forearm in mid-pronation. It is minimally active in slow flexion or with the forearm supine, but generates a powerful burst of activity in both flexion and extension when movement is rapid. Under these conditions, it develops a pronounced transarticular component of its force that helps to stabilize the elbow joint by balancing the centrifugal force of rapid swings in either direction.

**Testing** Brachioradialis can be seen and felt when the semi-pronated forearm is flexed against resistance.

### **Extensor carpi radialis longus**

**Attachments** Extensor carpi radialis longus arises from the distal third of the lateral supracondylar ridge of the humerus and the front of the lateral intermuscular septum (see Fig. 48.11A). Some fibres join from the common tendon of origin of the forearm extensors. The muscle belly ends at the junction of the proximal and middle thirds of the forearm in a flat tendon that runs along the lateral surface of the radius, deep to abductor pollicis longus and extensor pollicis brevis. The tendon passes deep to the dorsal carpal ligament in a radial groove behind the styloid process alongside extensor carpi radialis brevis. Just before it is inserted on the radial side of the dorsal surface of the base of the second metacarpal, the tendon is crossed obliquely by the tendon of extensor pollicis longus. It may send slips to the first or third metacarpal bones and contributes to the intermetacarpal ligaments.

**Relations** Extensor carpi radialis longus runs alongside brachioradialis and is partly overlapped by it.

**Vascular supply** Extensor carpi radialis longus receives its principal arterial supply from a single branch of the radial recurrent artery. Additional blood supply comes from branches of the radial collateral branch of the profunda brachii and directly from the radial artery in the distal part of the muscle.

**Innervation** Extensor carpi radialis longus is innervated by the radial nerve, C6 and 7.

**Actions** Extensor carpi radialis longus acts as an extensor and abductor of the wrist and midcarpal joints.

**Testing** The muscle belly and tendon of extensor carpi radialis longus can be palpated when the wrist is extended and abducted against resistance with the forearm pronated.

### **Extensor carpi radialis brevis**

**Attachments** Extensor carpi radialis brevis arises from the lateral epicondyle of the humerus by a common tendon shared with extensor digitorum, extensor digiti minimi and extensor carpi ulnaris, the radial collateral ligament of the elbow joint, a strong aponeurosis that covers

its surface, and adjacent intermuscular septa. The muscle belly ends at mid-forearm in a flat tendon that closely accompanies that of extensor carpi radialis longus to the wrist. The tendon passes under the extensor retinaculum, is crossed by the tendon of extensor pollicis longus, and is attached to the dorsal surface of the base of the third metacarpal on its radial side, distal to its styloid process, and on adjoining parts of the base of the second metacarpal. The tendons of both extensor carpi radialis longus and brevis share a common synovial sheath (see Fig. 50.41); they may split into slips that are variably attached to the second and third metacarpal bones. The muscles themselves may be united or may exchange muscular slips.

**Relations** Extensor carpi radialis brevis is shorter than extensor carpi radialis longus and is covered by it. The tendon passes deep to abductor pollicis longus and extensor pollicis brevis in the distal third of the forearm, then under the extensor retinaculum, where it lies in a shallow groove on the back of the radius, medial to the tendon of extensor carpi radialis longus, and separated from it by a low ridge.

**Vascular supply** Extensor carpi radialis brevis receives its arterial supply principally from two pedicles: a single branch from the radial recurrent artery, and a branch of the radial artery that arises about one-third of the way down the forearm. There is an additional blood supply proximally from branches from the radial collateral branch of the profunda brachii.

**Innervation** Extensor carpi radialis brevis is innervated by a branch from the radial nerve or from the posterior interosseous nerve, C7 and 8. The branch to extensor carpi radialis brevis arises before the nerve enters the arcade of Frohse.

**Actions** Extensor carpi radialis brevis is the prime dorsiflexor of the wrist. It acts with extensor carpi radialis longus as an extensor and abductor of the wrist and mid-carpal joints. It acts in synergism with the finger flexors when making a fist.

**Testing** The muscle belly and tendon of extensor carpi radialis brevis can be palpated when the wrist is extended and abducted against resistance with the forearm pronated.

### **'Mobile wad' of Henry**



Available with the *Gray's Anatomy* e-book

### **Extensor digitorum**

**Attachments** Extensor digitorum arises from the lateral epicondyle of the humerus via the common extensor tendon, the adjacent intermuscular septa and the antebrachial fascia (see Fig. 48.11A). The muscle belly divides in the distal third of the forearm into four tendons, which pass in a common synovial sheath with the tendon of extensor indicis through a tunnel under the extensor retinaculum (see Fig. 50.41). The tendons diverge on the dorsum of the hand, one to each finger. The tendon to the index finger is accompanied by extensor indicis, which lies ulnar (medial) to it. On the dorsum of the hand, adjacent tendons are linked by three variable intertendinous connections (juncturae tendinae), which are inclined distally and radially (see Figs 49.23, 50.6). The digital attachments enter a fibrous expansion on the dorsum of the proximal phalanges, to which lumbrical, interosseous and digital extensor tendons all contribute.

The tendons of extensor digitorum may be variably deficient. More commonly, they are doubled, or even tripled, in one or more digits: most often, the index finger or the middle finger. Occasionally, a slip of tendon passes to the thumb. The arrangement of the intertendinous connections on the dorsum of the hand is highly variable. The medial connection is strong and pulls the tendon of the little finger towards that of the ring finger, whereas the connection between the middle two tendons is weak and may be absent.

**Relations** The extensor tendon to the index finger lies radial (lateral) to the tendon of extensor indicis. The extensor tendon to the little finger lies radial (lateral) to the tendon of extensor digiti minimi.

**Vascular supply** The proximal third of extensor digitorum is supplied by branches from the radial recurrent artery, and the distal two-thirds are supplied by branches from the posterior interosseous artery. The very distal portion is supplied by a perforating branch from the anterior interosseous artery that passes through the interosseous membrane.



Henry (1957) described a 'mobile wad' of three muscles, brachioradialis and the extensors carpi radialis longus and brevis, which are visible and palpable below the lateral epicondyle of the humerus, lying on the lateral aspect of the radius. The wad is tethered by a fan-shaped leash of vessels derived from the radial artery (the radial recurrent artery) that has to be released before the muscles can be fully mobilized. Its anterior and posterior margins are used to define the surgical approaches to the radius and to discover planes of cleavage. The anterior margin is used to define the anterior approach (of Henry) to the radius and can be extended down to the wrist at the level of the insertion of brachioradialis. This requires detachment of supinator from the radius, taking care not to damage the posterior interosseous nerve. More distally, pronator teres is released. This approach is particularly useful for plating fractures of the radius. The posterior margin is used to expose the back of the forearm (posterior approach of Thompson) and, if need be, the posterior interosseous nerve, which, in any event, should be protected; unlike the anterior approach, this approach cannot be extended.

**Innervation** Extensor digitorum is innervated by the posterior interosseous nerve, C7 and 8.

**Actions** Extensor digitorum can extend any or all of the joints over which it passes: wrist, metacarpophalangeal, and proximal and distal interphalangeal joints (the latter two via the extensor expansion of the digits). When acting on the metacarpophalangeal joints, extensor digitorum tends to spread the digits apart because of the different axes of the individual joints imposed by the transverse arch of the hand. The intertendinous bands stabilize the individual tendons as they cross the dorsum of the hand and permit some extension when a tendon has been divided in the metacarpal segment. The extent to which this occurs depends on the presence and orientation of the juncturae tendinae.

**Testing** The tendons of extensor digitorum can be readily felt, and usually seen, when the fingers are extended against resistance and the forearm is pronated.

### Extensor digiti minimi

**Attachments** Extensor digiti minimi is medial to, and usually covered by, extensor digitorum; proximally, the bellies of these two muscles may be connected and indistinct from each other, whereas, in the mid-forearm, extensor digiti minimi is clearly discernible as a slender muscle (Fig. 49.23). It arises from the common extensor tendon by a thin, tendinous slip and adjacent intermuscular septa, and frequently has an additional origin from the antebrachial fascia. It forms a long tendon that constitutes the fifth dorsal compartment as it proceeds beneath the extensor retinaculum. Distal to the retinaculum (see Fig. 50.41), the tendon typically splits into two, and the lateral slip is joined by a tendon from extensor digitorum. All three tendons are attached to the dorsal digital expansion of the fifth digit; there may be a slip to the fourth digit. Extensor digiti minimi is rarely absent but, sometimes, it is fused with extensor digitorum.

**Relations** The tendon of extensor digiti minimi lies ulnar (medial) to the common extensor tendon to the little finger. The dorsal branch of the ulnar artery and the dorsal branch of the ulnar nerve pass superficial to the extensor retinaculum and cross the tendon of extensor digiti minimi as it lies beneath the retinaculum.

**Vascular supply** Extensor digiti minimi is supplied by branches from the radial recurrent and posterior interosseous arteries, and by terminal branches from the anterior interosseous artery after it has pierced the interosseous membrane.

**Innervation** Extensor digiti minimi is innervated by the posterior interosseous nerve, C7 and 8.

**Actions** Extensor digiti minimi can extend any of the joints of the little finger, or contribute to wrist extension. It permits extension of the little finger independently of the other digits, even in extremes of ulnar or radial wrist deviation.

**Testing** Extensor digiti minimi is tested by extending the little finger while holding the remaining fingers flexed at the metacarpophalangeal joints; this eliminates any simultaneous contraction of extensor digitorum.

### Extensor carpi ulnaris

**Attachments** Extensor carpi ulnaris arises from the lateral epicondyle (where it forms the most medial part of the common extensor origin), the posterior border of the ulna (by an aponeurosis shared with flexor carpi ulnaris and flexor digitorum profundus) and overlying fascia (see Fig. 49.6). It runs between extensor digiti minimi and anconeus proximally, and extensor digiti minimi and flexor carpi ulnaris in the mid-forearm (see Figs 49.31, 49.32). It forms a tendon in the distal third of the forearm that slides in a groove between the head and the styloid process of the ulna. It forms the content of the sixth dorsal compartment as it runs under the extensor retinaculum (see Fig. 50.41) and is attached to a tubercle on the medial side of the base of the fifth metacarpal.

**Vascular supply** Proximally, extensor carpi ulnaris receives branches from the radial recurrent artery. Distally, it is supplied by several branches from the posterior interosseous artery.

**Innervation** Extensor carpi ulnaris is innervated by the posterior interosseous nerve, C7 and 8.

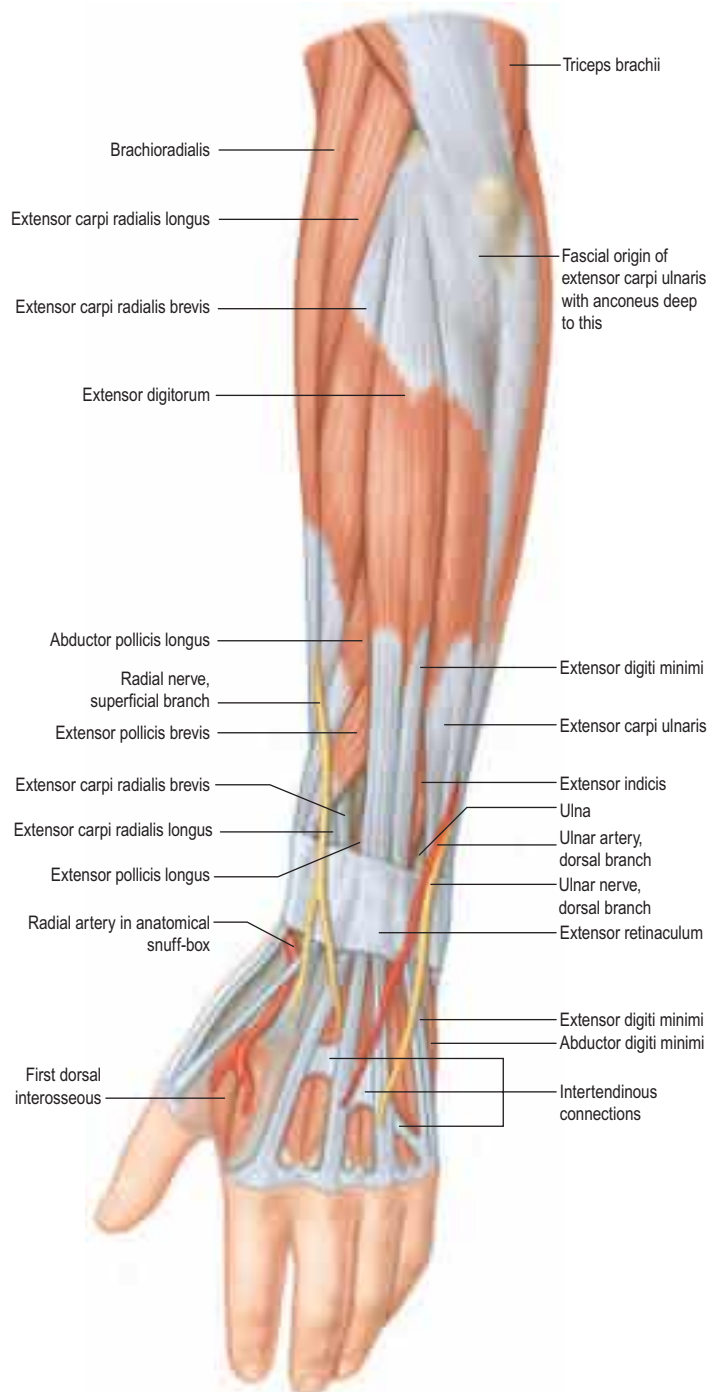


Fig. 49.23 The superficial extensor muscles of the left forearm.

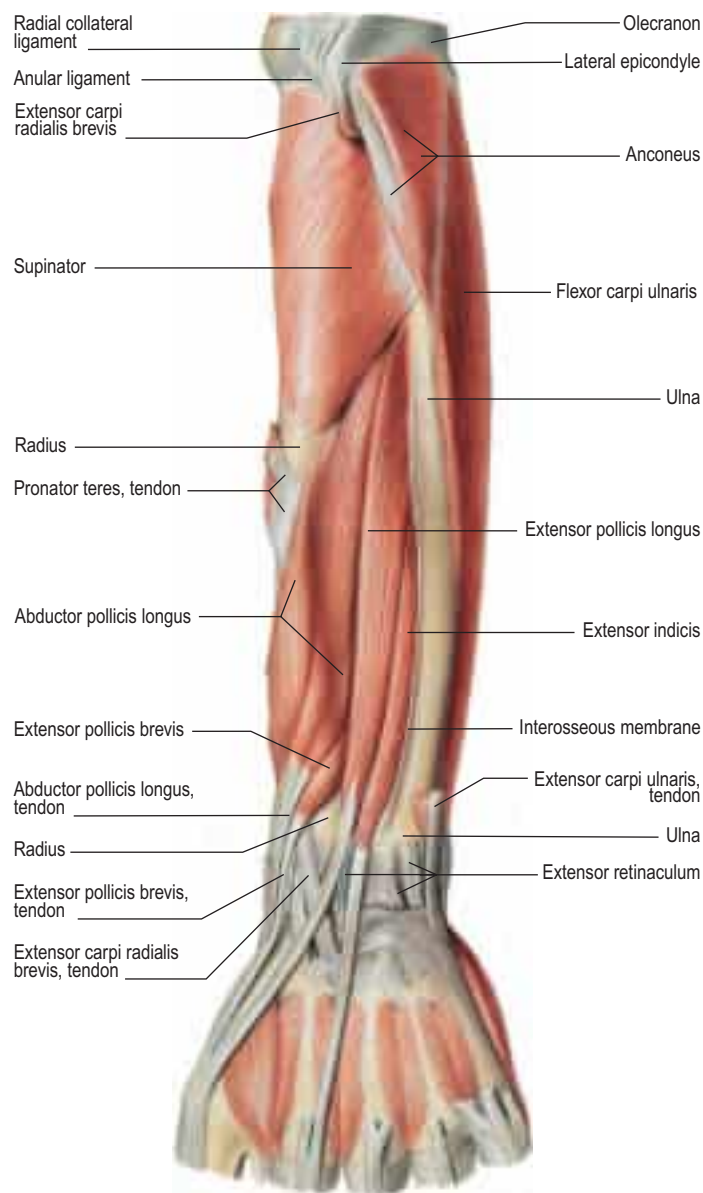
**Actions** Extensor carpi ulnaris works synergistically with extensors carpi radialis longus and brevis and the digital flexors to effect balanced wrist extension, enabling a power grip with a clenched fist. It also acts with flexor carpi ulnaris to adduct the hand.

**Testing** The tendon can be felt lateral to the groove that overlies the posterior subcutaneous border of the ulna when the wrist is adducted against resistance.

### Anconeus

**Attachments** Anconeus is a short, triangular muscle posterior to the elbow joint, partially and variably blended with triceps or extensor carpi ulnaris (Fig. 49.24). It arises by a separate tendon from the posterior surface of the lateral epicondyle of the humerus, proximal to the common extensor tendon. The tendon lies beneath the muscle belly of extensor carpi radialis longus. The muscle fibres run obliquely and medially towards the ulna, covering the posterior aspect of the annular ligament, and are attached to the lateral aspect of the olecranon and proximal quarter of the posterior surface of the shaft of the ulna.

**Vascular supply** Anconeus is supplied by branches of the posterior interosseous recurrent artery. A small number of musculocutaneous perforators reach the skin overlying the muscle.



**Fig. 49.24** The deep layer of the dorsal muscles of the forearm. (With permission from Waschke J, Paulsen F (eds), Sobotta Atlas of Human Anatomy, 15th ed, Elsevier, Urban & Fischer. Copyright 2013.)

**Innervation** Anconeus is innervated by the radial nerve, C6, 7 and 8.

**Actions** Anconeus is an extensor of the elbow. Its principal function is not fully understood but it may control ulnar abduction in pronation, which is necessary if the forearm is to turn over the hand without translating it medially. In this way, a tool can be revolved 'on the spot' or it can be swept through an arc.

## Deep extensor compartment

The five deep forearm extensor muscles are abductor pollicis longus, extensor pollicis longus and extensor pollicis brevis (all of which act on the thumb), extensor indicis and supinator. Apart from supinator, the other four deep forearm extensors all arise from the bones of the forearm and do not, therefore, act on the elbow.

### Abductor pollicis longus

**Attachments** Abductor pollicis longus arises in part from the posterior surface of the shaft of the ulna, just distal to the insertion of anconeus, and from the adjoining interosseous membrane (see Fig. 49.6). The muscle runs obliquely in a lateral and distal direction from an apex formed by anconeus and supinator, and is joined by fibres that arise from the middle third of the posterior surface of the radius distal to the attachment of supinator (see Fig. 49.24). As it proceeds distally, abductor pollicis longus becomes more superficial in the forearm, where it is visible as an oblique elevation. The muscle fibres form a tendon just proximal to the wrist. The tendon runs in a groove on the lateral side of the distal end of the radius, accompanied by the tendon

of extensor pollicis brevis. It runs under the extensor retinaculum in the first dorsal compartment (see Fig. 50.41). It usually splits into two slips; one is attached to the radial side of the first metacarpal base, and the other is attached to the trapezium. Slips from the tendon may continue into opponens pollicis or abductor pollicis brevis. Occasionally, the muscle itself may be wholly or partially divided.

**Relations** Abductor pollicis longus runs on the radial aspect of extensor pollicis longus. The posterior interosseous nerve runs over the belly of abductor pollicis longus with the accompanying posterior interosseous artery; the nerve runs to the border of abductor pollicis longus and then dives between the muscles bellies of abductor pollicis longus and extensor pollicis longus at the apex of their divergence.

**Vascular supply** Abductor pollicis longus is supplied proximally by a lateral branch from the posterior interosseous artery. Distally, it is supplied on the medial side by a perforating branch from the anterior interosseous artery.

**Innervation** Abductor pollicis longus is innervated by the posterior interosseous nerve, C7 and 8.

**Actions** Abductor pollicis longus abducts the metacarpal of the thumb. It is the prime radial abductor of the thumb. It also acts to abduct the wrist. With abductor pollicis brevis, it effects palmar abduction.

**Testing** The tendon of abductor pollicis longus can be seen and felt at the radial aspect of the anatomical snuff-box when the thumb and wrist are abducted against resistance at the carpometacarpal joint.

### Extensor pollicis longus

**Attachments** Extensor pollicis longus arises from the lateral part of the middle third of the posterior surface of the shaft of the ulna below abductor pollicis longus and from the adjacent interosseous membrane. It partly overlies the proximal attachment of the smaller extensor pollicis brevis (see Fig. 49.24). The tendon is formed just proximal to the wrist and passes through the third compartment of the extensor retinaculum in a narrow, oblique groove on the back of the distal end of the radius (see Fig. 50.41). It turns around a bony fulcrum, Lister's tubercle, which changes its line of pull from that of the forearm to that of the thumb, and is attached to the base of the distal phalanx of the thumb. The sides of the tendon are joined on the dorsum of the proximal phalanx by expansions from the tendon of abductor pollicis brevis laterally and from the first palmar interosseous and adductor pollicis medially.

**Relations** Extensor pollicis longus runs between abductor pollicis longus and extensor indicis in the forearm. The posterior interosseous artery runs over its muscle belly. After passing around Lister's tubercle, the tendon of extensor pollicis longus crosses the tendons of extensor carpi radialis brevis and longus obliquely.

**Vascular supply** Extensor pollicis longus is supplied on its superficial surface by branches from the posterior interosseous artery, and on its deep surface by perforating branches from the anterior interosseous artery.

**Innervation** Extensor pollicis longus is innervated by the posterior interosseous nerve, C7 and 8.

**Actions** Extensor pollicis longus extends the distal phalanx of the thumb. Acting in association with extensor pollicis brevis and abductor pollicis longus, it extends the proximal phalanx and the metacarpal. In continued action, as a consequence of the obliquity of its tendon, extensor pollicis longus adducts the extended thumb and rotates it laterally. It is a weak extensor of the wrist.

**Testing** The tendon of extensor pollicis longus can be palpated at the ulnar border of the anatomical snuff-box when the thumb is extended at the interphalangeal joint against resistance. When the thumb is opposed and adducted, abductor pollicis brevis can extend the interphalangeal joint of the thumb, mimicking the action of extensor pollicis longus.

### Extensor pollicis brevis

**Attachments** Extensor pollicis brevis arises from the posterior surface of the radius distal to abductor pollicis longus and from the adjacent interosseous membrane (see Fig. 49.6). It runs between abductor



pollicis longus and extensor pollicis longus. The tendon is attached to the base of the proximal phalanx of the thumb and, commonly, has an additional attachment to the base of the distal phalanx, usually through a fasciculus that joins the tendon of extensor pollicis longus. Extensor pollicis brevis may be absent or fused completely with abductor pollicis longus.

**Relations** Extensor pollicis brevis is ulnar (medial) to, and closely connected with, abductor pollicis longus. In the distal forearm, the two muscles emerge between extensor carpi radialis brevis and extensor digitorum, and pass obliquely across the tendons of extensors carpi radialis brevis and longus (see Fig. 49.24). They cover the distal part of brachioradialis, and pass through the most lateral compartment of the extensor retinaculum in a single synovial sheath, sharing a groove in the distal radius. Ultimately, they cross, superficial to the radial styloid process and radial artery, to reach the dorsolateral base of the proximal phalanx of the thumb.

**Vascular supply** Extensor pollicis brevis is supplied from branches from the posterior interosseous artery, together with perforating branches from the anterior interosseous artery.

**Innervation** Extensor pollicis brevis is innervated by the posterior interosseous nerve, C7 and 8.

**Actions** Extensor pollicis brevis extends the proximal phalanx and metacarpal of the thumb.

**Testing** The tendon of extensor pollicis brevis can be felt at the radial border of the anatomical snuff-box, lying medial to the tendon of abductor pollicis longus, when the metacarpophalangeal joint of the thumb is extended against resistance.

### Extensor indicis

**Attachments** Extensor indicis is a narrow muscle that lies medial and parallel to extensor pollicis longus (see Fig. 49.24). It arises from the posterior surface of the ulna distal to extensor pollicis longus and the adjacent interosseous membrane. The tendon is formed just proximal to the wrist and passes under the extensor retinaculum in a common compartment with the tendons of extensor digitorum (see Fig. 50.41). Opposite the head of the second metacarpal, it joins the ulnar side of the tendon of extensor digitorum that serves the index finger. Extensor indicis occasionally sends accessory slips to the extensor tendons of other digits. Its tendon may be interrupted on the dorsum of the hand by an additional muscle belly (extensor indicis brevis manus).

**Relations** The posterior interosseous artery passes over the muscle belly of extensor indicis. On the dorsum of the hand, the tendon of extensor indicis lies on the ulnar aspect of the tendon of extensor digitorum to the index finger.

**Vascular supply** Extensor indicis is supplied on its superficial surface by branches from the posterior interosseous artery, and on its deep surface by perforating branches from the anterior interosseous artery.

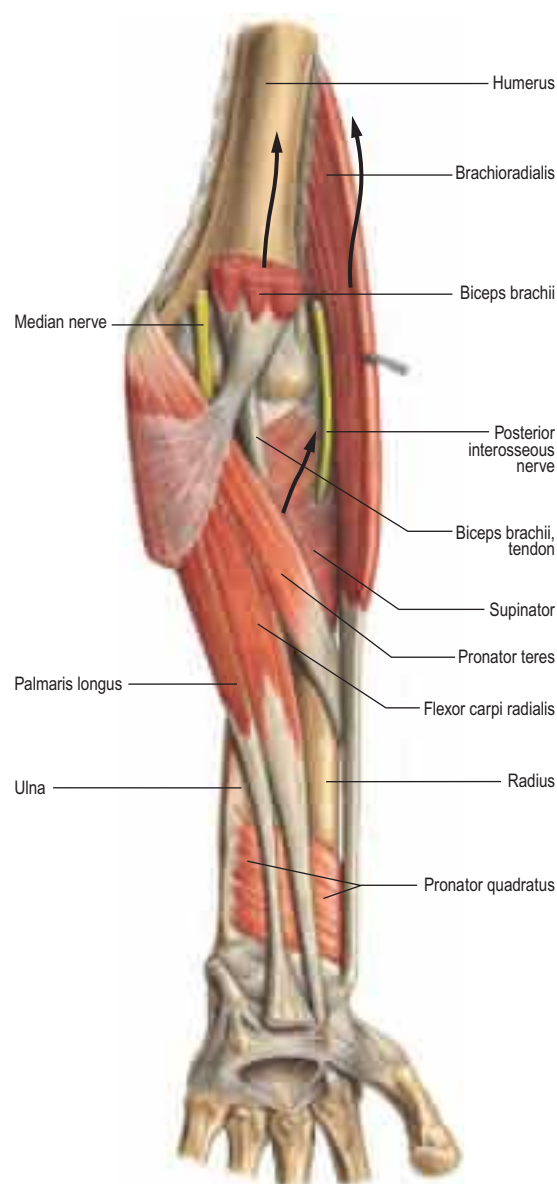
**Innervation** Extensor indicis is innervated by the posterior interosseous nerve, C7 and 8.

**Actions** Extensor indicis extends the index finger independently of the other digits. It is a weak extensor of the wrist.

**Testing** Extensor indicis is tested by extending the index finger while holding the remaining fingers flexed at the metacarpophalangeal joints in order to eliminate the effects of any simultaneous contraction of extensor digitorum.

### Supinator

**Attachments** Supinator is a wide muscle that surrounds the proximal third of the radius. It has superficial and deep layers. The two parts arise together – the superficial by tendinous fibres, and the deep by muscular fibres – from the lateral epicondyle of the humerus, the radial collateral ligament of the elbow joint, the anular ligament of the superior radio-ulnar joint, the supinator crest of the ulna and the posterior part of the triangular depression in front of it, and an aponeurosis that covers the muscle (see Figs 49.5, 49.6; Figs 49.25–49.26). Supinator is attached distally to the lateral surface of the proximal third of the radius, down to the insertion of pronator teres. The radial attachment extends on to the anterior and posterior surfaces between the anterior oblique line and the fainter posterior oblique ‘ridge’. Supinator is subject to frequent



**Fig. 49.25** The forearm, antebrachium, in supination position. (With permission from Waschke J, Paulsen F (eds), Sobotta Atlas of Human Anatomy, 15th ed, Elsevier, Urban & Fischer. Copyright 2013.)

variation; small parts have acquired individual names, e.g. lateral and medial tensors of the anular ligament.

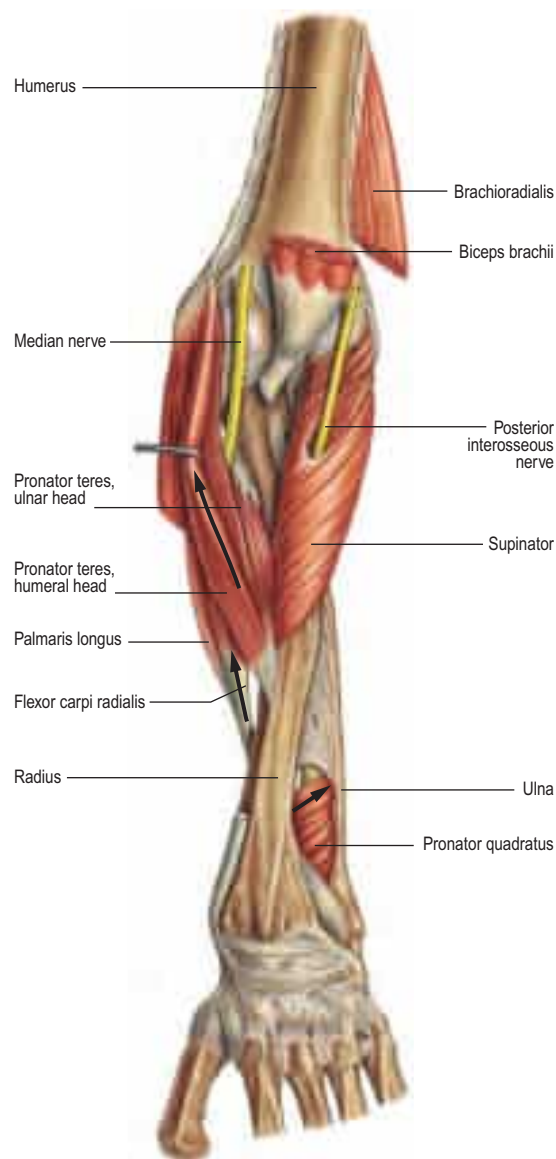
**Relations** Supinator emerges under the lower border of anconeus and is related to abductor pollicis longus at its inferior medial border. Extensor carpi radialis brevis and longus pass over its radial aspect. The posterior interosseous nerve enters the forearm by passing between its superficial and deep heads.

**Vascular supply** The superficial part of supinator is supplied by branches from the radial recurrent artery. The deep part of supinator is supplied by branches from the posterior interosseous artery and the posterior interosseous recurrent artery.

**Innervation** Supinator is innervated by the posterior interosseous nerve, C6 and 7.

**Actions** Supinator rotates the radius to bring the palm to face anteriorly. It acts alone in slow, unopposed supination, and together with biceps brachii in fast or forceful supination. An object, which potentially may be heavy, is often picked up with the forearm initially pronated. The more powerful supinator lifts the object against gravity; rotation is often combined with increasing elbow flexion in order to bring the object towards the eyes.

**Testing** Supinator lies too deep to be palpated. It may be assessed by supinating the forearm against resistance with the elbow extended, on



**Fig. 49.26** The forearm, antebrachium, in pronation position. (With permission from Waschke J, Paulsen F (eds), *Sobotta Atlas of Human Anatomy*, 15th ed, Elsevier, Urban & Fischer. Copyright 2013.)

the grounds that biceps brachii is inactive on supination when the elbow is fully extended.

## COMPARTMENT SYNDROME OF THE FOREARM



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## VASCULAR SUPPLY AND LYMPHATIC DRAINAGE

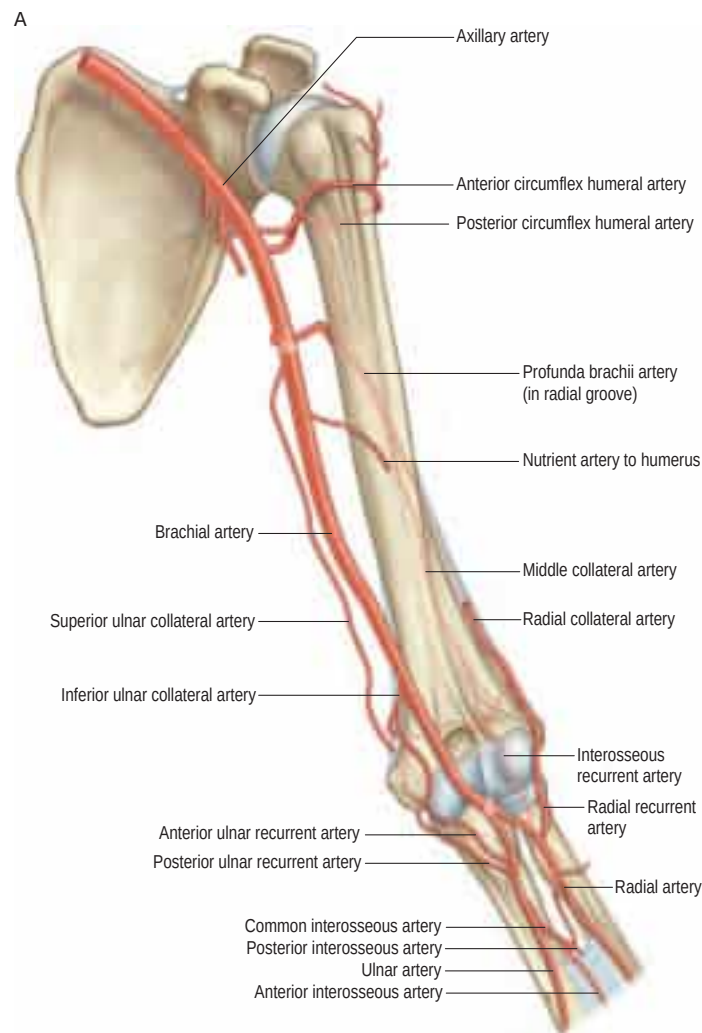
### ARTERIES

#### Brachial artery

The brachial artery is medial in the arm but moves towards the midline, medial to the tendon of biceps brachii and deep to the bicipital aponeurosis. The latter separates the artery from the anterior structures, namely: the skin, superficial fascia and the median cubital vein. Posteriorly, brachialis separates it from the elbow joint. The median nerve passes close to the medial side of the artery. The artery divides into its terminal branches, the radial and ulnar arteries, at the level of the radial neck (Fig. 49.27A).

#### Radial artery

The radial artery is smaller than the ulnar artery, yet appears a more direct continuation of the brachial artery (see Figs 49.21–49.22). It



**Fig. 49.27** **A**, The arterial anastomoses around the left elbow joint, viewed from the anterior aspect. **B**, Arteriogram demonstrating anterior arterial anastomoses around the elbow.

passes deep to brachioradialis and gives off the radial recurrent artery before continuing into the forearm. It descends along the lateral side of the forearm, accompanied by paired venae comitantes, from the medial side of the neck of the radius to the wrist. The artery is medial to the radial shaft proximally, and anterior to it distally. Proximally, it is overlapped anteriorly by the belly of brachioradialis, but elsewhere in its course it is covered only by the skin and superficial and deep fasciae. Its posterior relations in the forearm are, successively, the tendon of biceps, supinator, the distal attachment of pronator teres, the radial head of flexor digitorum superficialis, flexor pollicis longus, pronator quadratus and the lower end of the radius (where its pulsation is

A compartment syndrome exists where increased pressure in an osseofascial compartment leads to decreased perfusion of the contents of the compartment. In the forearm, this commonly occurs as a result of trauma, external pressure, burns, arterial injury or post-ischaemic swelling. The resultant increase in interstitial pressure causes vascular occlusion and myoneural ischaemia. The patient presents with pain that is disproportionate to the severity of the injury and which is provoked or markedly increased by passive extension of the fingers. There may also be reduced sensation or paraesthesiae in the affected hand. If a compartment syndrome is suspected, compartment pressure measurements may be indicated, but, in unequivocal cases, the definitive treatment is emergency fasciotomies of the forearm. The absolute intracompartmental pressure threshold for decompression is the subject of debate: an absolute value of 30 mmHg is generally accepted, although some authorities prefer a figure that comes within 30 mmHg of diastolic pressure. The pressure in each compartment should be measured separately.

Incisions to decompress the flexor compartment ([Ronel et al 2004](#)) run from the medial epicondyle and curve down the anterior aspect of the forearm, ending on the radial side of flexor carpi ulnaris; from here, they may be extended distally to decompress the carpal tunnel, if necessary. The bicipital aponeurosis (lacertus fibrosus) is routinely released to decompress the median nerve. After retraction of the superficial flexors, the fascia over the deep muscles of the forearm may be incised over its full length. Both median and ulnar nerves should be inspected to ensure that their decompression is complete. The mobile wad and dorsal compartment are decompressed through a straight longitudinal incision that starts just distal to the lateral epicondyle; this is deepened through the plane between extensor digitorum and extensor carpi radialis brevis, and ends over the midline of the wrist. In both cases, the wounds are left open and dressed, and only closed when circumstances permit.

Failure to decompress a compartment syndrome promptly may lead to Volkmann's ischaemic contracture with severe loss of function in the affected forearm and hand. Subsequent amputation is not unknown.



most accessible). Brachioradialis is lateral to the artery throughout its length. Pronator teres is medial to the proximal part of the artery, and the tendon of flexor carpi radialis is medial to the distal portion. The superficial radial nerve lies lateral to the middle third of the radial artery; multiple branches from the artery supply the nerve throughout its length.

The radial artery may occasionally arise from a continuation of a superficial brachial artery, or as a high proximal division of an otherwise normal brachial artery. It can give rise to the common interosseous artery.

### Branches in the forearm

**Radial recurrent artery** The radial recurrent artery is given off by the radial artery just distal to the elbow joint (see Figs 49.21–49.22, 49.27). It passes between the superficial radial nerve and the posterior interosseous nerve before ascending beneath brachioradialis, anterior to supinator and brachialis. It supplies these muscles and then anastomoses with the radial collateral branch of the profunda brachii artery. The radial recurrent artery also gives branches to extensor carpi radialis longus, extensor digitorum, extensor digiti minimi and extensor carpi ulnaris. Its cutaneous branches are described on page 837.

**Muscular branches** Muscular branches are distributed to the muscles on the radial side of the forearm. Direct branches from the radial artery supply the radial insertion of pronator teres, the antero-lateral aspect of flexor carpi radialis, the lateral surface of flexor digitorum superficialis, the lateral half of flexor pollicis longus, and the distal sections of extensor carpi radialis longus and extensor carpi radialis brevis.

### Ulnar artery

The ulnar artery arises from the brachial artery just distal to the flexion crease of the elbow at the level of the radial neck, and reaches the medial side of the forearm in its upper third (see Figs 49.21–49.22). It is larger than the radial artery and is accompanied throughout its length by venae comitantes. In the forearm, the ulnar artery initially lies on brachialis before passing deep to pronator teres, flexor carpi radialis, palmaris longus and flexor digitorum superficialis. It then lies on flexor digitorum profundus, between flexor carpi ulnaris and flexor digitorum superficialis, and is covered by the skin and superficial and deep fasciae. The artery crosses the flexor retinaculum, lateral to the ulnar nerve and pisiform bone, to enter the hand. The median nerve is a medial relation for approximately 2.5 cm distal to the elbow, and then crosses the artery, from which it is separated by the ulnar head of pronator teres. The ulnar nerve lies medial to the distal two-thirds of the artery, which supplies the nerve throughout its length; the palmar cutaneous branch of the ulnar nerve descends along the ulnar artery to reach the hand.

The ulnar artery may arise above the elbow, when it can lie superficial to the forearm flexors under the deep fascia; only rarely is it subcutaneous. When this occurs, the brachial artery supplies the common interosseous and the ulnar recurrent arteries.

### Branches in the forearm

**Anterior ulnar recurrent artery** The anterior ulnar recurrent artery arises from the ulnar artery distal to the elbow and close to the origin of the ulnar artery itself. It ascends between brachialis and pronator teres, which it supplies, then anastomoses with the inferior ulnar collateral artery anterior to the medial epicondyle (see Fig. 49.27A).

**Posterior ulnar recurrent artery** The posterior ulnar recurrent artery arises from the ulnar artery distal to the anterior ulnar recurrent artery, passes dorsomedially between flexor digitorum profundus and flexor digitorum superficialis, and ascends between the posterior aspect of the medial epicondyle and the olecranon. It is deep to flexor carpi ulnaris, ascending between its heads with the ulnar nerve. It supplies flexor digitorum profundus, flexor digitorum superficialis, flexor carpi ulnaris, the ulnar nerve, underlying bone and the elbow joint, and anastomoses with the ulnar collateral artery and interosseous recurrent artery. The posterior ulnar recurrent artery and ulnar nerve are both vulnerable to injury behind the medial epicondyle (Figs 49.28–49.29).

**Common interosseous artery** The common interosseous artery is a short branch of the ulnar artery (see Figs 49.10, 49.22). It arises just distal to the radial tuberosity and passes back to the proximal border of the interosseous membrane, where it divides into the anterior and posterior interosseous arteries. Occasionally, the common interosseous artery is a branch of the radial artery.

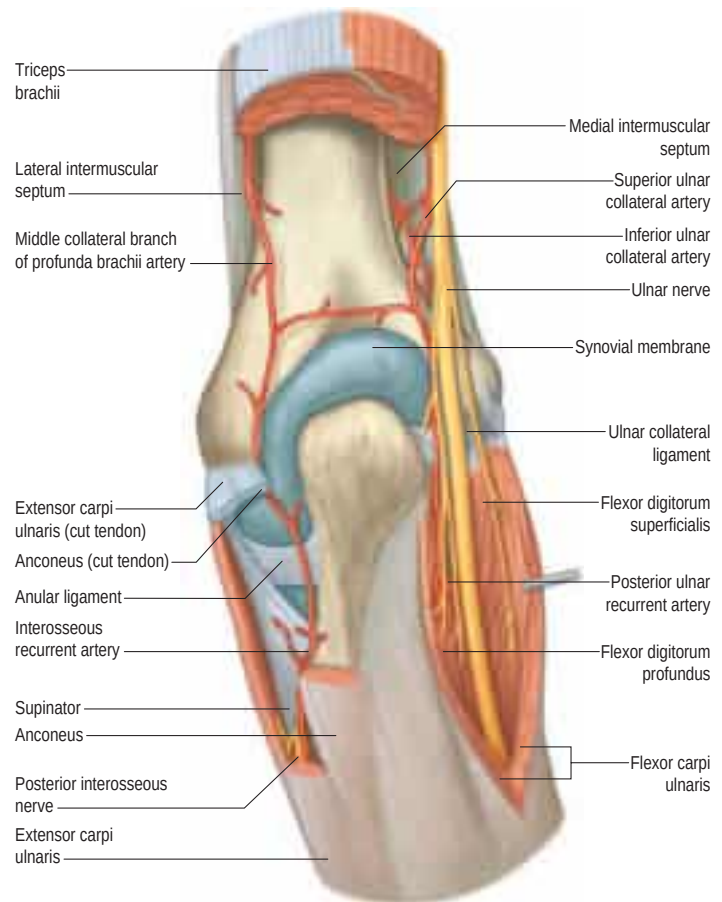


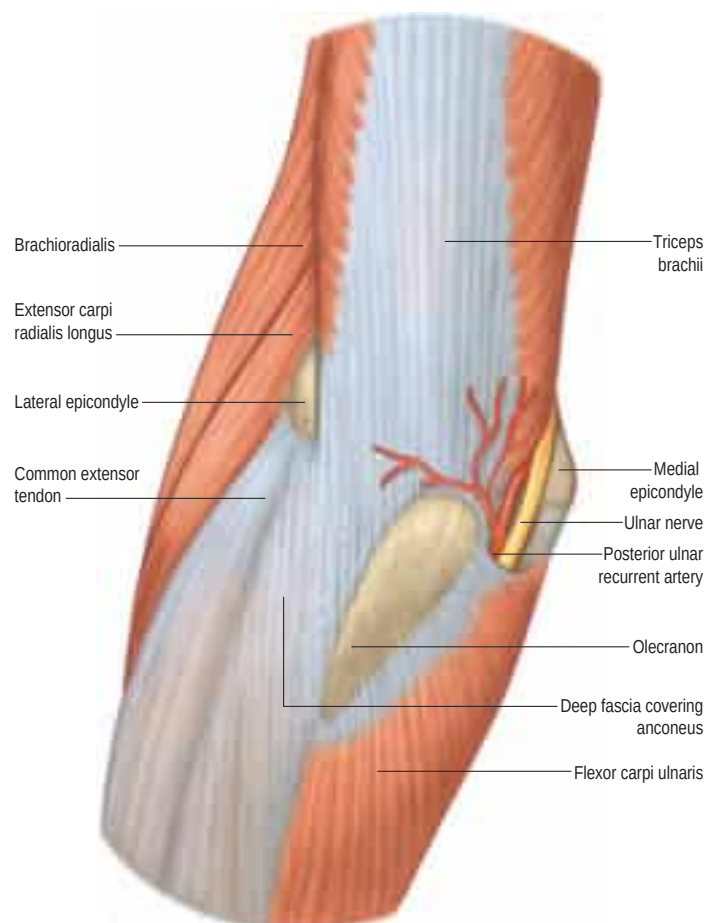
Fig. 49.29 The posterior aspect of the left elbow region: deep structures.

**Anterior interosseous artery** The anterior interosseous artery arises from the common interosseous artery and descends on the anterior aspect of the interosseous membrane with the anterior interosseous branch of the median nerve (see Figs 49.10, 49.22, 49.27, 49.32). It is overlapped by contiguous sides of flexor digitorum profundus and flexor pollicis longus. Shortly after its origin, it usually gives off a slender median artery. This accompanies and supplies the median nerve as far as the palm, where it may join the superficial palmar arch or end as one or two palmar digital arteries. The median artery can also arise from the ulnar or the common interosseous artery. Muscular and nutrient branches from the anterior interosseous artery pierce the interosseous membrane to supply deep extensor muscles and the radius and ulna, respectively. A branch descends deep to pronator quadratus before piercing the interosseous membrane to join the anterior 'carpal' arch.

The anterior interosseous artery leaves the anterior compartment by piercing the interosseous membrane proximal to pronator quadratus. It anastomoses with the posterior interosseous artery in the posterior compartment of the forearm, and travels through a tunnel under the extensor retinaculum with the tendons of the digital extensors before joining the dorsal carpal arch. Three small cutaneous perforating branches supply the skin over the lower lateral border of the forearm.

**Posterior interosseous artery** The posterior interosseous artery arises from the common interosseous artery and is usually smaller than the anterior interosseous artery (see Figs 49.10, 49.27A). It passes dorsally between the oblique cord and proximal border of the interosseous membrane, and then between supinator and abductor pollicis longus. It descends deep in the groove between extensor carpi ulnaris and the extensor digiti minimi component of extensor digitorum. While in the groove, it gives rise to multiple muscular branches that supply these muscles and fasciocutaneous perforators that travel in the intermuscular septum between extensor carpi ulnaris and extensor digiti minimi. The posterior interosseous artery accompanies the deep branch of the radial nerve (posterior interosseous nerve) on abductor pollicis longus. Distally, it anastomoses with the terminal part of the anterior interosseous artery and the dorsal carpal arch.

Sometimes, the posterior interosseous artery disappears halfway down the forearm, in which case the anterior interosseous artery pierces the interosseous membrane more proximally to anastomose with it. Occasionally, the two vessels do not anastomose.



**Fig. 49.28** The posterior aspect of the left elbow: superficial structures.

**Posterior interosseous recurrent artery** The posterior interosseous recurrent artery leaves the posterior interosseous artery near its origin and ascends between the lateral epicondyle and olecranon, either on or through supinator, and deep to anconeus. It anastomoses with the middle collateral branch of the profunda brachii artery, posterior ulnar recurrent artery and ulnar collateral artery. Occasionally, the artery is absent. Its cutaneous branches are described on [page 837](#).

## Collateral circulation at the elbow



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## VEINS

### Deep veins

Venae comitantes run with the radial and ulnar arteries to drain the deep and superficial palmar venous arches, respectively. They unite near the elbow as paired brachial veins. The radial veins are smaller and receive the deep dorsal veins of the hand. The ulnar veins are larger and drain the deep palmar venous arch. They connect with superficial veins near the wrist and receive the venae comitantes of the anterior and posterior interosseous arteries near the elbow. A large branch connects them to the median cubital vein.

### Superficial veins

#### Cephalic vein

The cephalic vein usually forms over the anatomical snuff-box on the radial side of the wrist from the radial end of the dorsal venous plexus (see [Figs 46.4, 50.39](#)). It curves proximally around the radial side of the forearm, receiving tributaries from both radial and ulnar aspects of the forearm. Distal to the elbow, a branch, the median cubital vein, diverges proximomedially to reach the basilic vein. The median cubital vein is joined by a branch from the deep veins ([Fig. 49.30](#)). The further course of the cephalic vein is described on [page 830](#).

**Accessory cephalic vein** The accessory cephalic vein may arise either from a dorsal forearm plexus or from the ulnar side of the dorsal venous network in the hand. It joins the cephalic vein distal to the elbow. A large oblique vein often connects the basilic and cephalic veins dorsally in the forearm.

#### Basilic vein

The basilic vein arises medially in the dorsal venous network of the hand (see [Figs 46.4, 50.39](#)). It ascends posteromedially in the forearm, inclining forwards to the anterior surface distal to the elbow, where it is joined by the median cubital vein. It then ascends superficially between biceps and pronator teres, and is crossed by filaments of the medial cutaneous nerve of the forearm that pass both superficial and deep to the vein. The basilic vein continues as the axillary vein from the lower border of teres major ([p. 830](#)).

#### Median vein

The median vein of the forearm drains the superficial palmar venous plexus and ascends through the anterior part of the forearm to join either the basilic or the median cubital vein (see [Figs 48.34, 49.30](#)). It may divide distal to the elbow to join both veins.

## LYMPHATIC DRAINAGE

One or two supratrochlear nodes are superficial to the deep fascia proximal to the medial epicondyle and medial to the basilic vein; their efferents accompany the vein to join the deep lymph vessels. Small, isolated nodes sometimes occur along the radial, ulnar and interosseous vessels, in the cubital fossa near the bifurcation of the brachial artery, or in the arm medial to the brachial vessels.

## INNERVATION

### Median nerve

In the cubital fossa, the median nerve lies medial to the brachial artery, deep to the bicipital aponeurosis and anterior to brachialis. It usually enters the forearm between the heads of pronator teres, although, occasionally, it may pass posterior to both heads of the muscle, or it may pass through its humeral head ([Fig. 49.31](#); see [Figs 49.21–49.22](#)). The nerve crosses to the lateral side of the ulnar artery, from which it is separated by the deep head of pronator teres, passes behind a tendinous bridge between the humero-ulnar and radial heads of flexor digitorum superficialis, and descends through the forearm posterior and adherent to flexor digitorum superficialis and anterior to flexor digitorum profundus. It is accompanied by the median branch of the anterior interosseous artery. The first branch of the median nerve to pronator teres arises 2–3 cm above the medial epicondyle. A second branch, or leash of branches, arises at the level of the tip of the medial epicondyle and innervates palmaris longus, flexor carpi radialis and flexor digitorum superficialis. About 5 cm proximal to the flexor retinaculum, the median nerve emerges from behind the lateral edge of flexor digitorum superficialis, and becomes superficial just proximal to the wrist. Here, it lies between the tendons of flexor digitorum superficialis and flexor carpi radialis, projecting laterally from beneath the tendon of palmaris longus. It then passes deep to the flexor retinaculum into the palm (see [Fig. 50.41](#)).

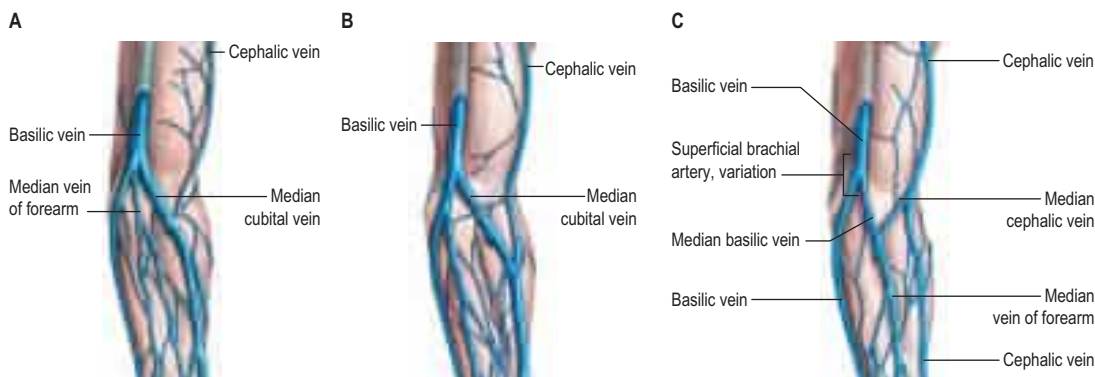
### Branches in the forearm

#### Anterior interosseous nerve

The anterior interosseous nerve arises from the posterior aspect of the median nerve between the two heads of pronator teres, just distal to the branches to the superficial forearm flexors and proximal tendinous arch of flexor digitorum superficialis (see [Figs 49.10, 49.22](#)). It descends with the anterior interosseous artery anterior to the interosseous membrane, between and deep to flexor pollicis longus and flexor digitorum profundus ([Fig. 49.32](#)), supplying flexor pollicis longus and the lateral part of flexor digitorum profundus (to the index and middle fingers). Terminally, the anterior interosseous nerve lies posterior to pronator quadratus, which it supplies via its deep surface. It also supplies articular branches to the distal radio-ulnar, radiocarpal and carpal joints.

#### Palmar cutaneous branch

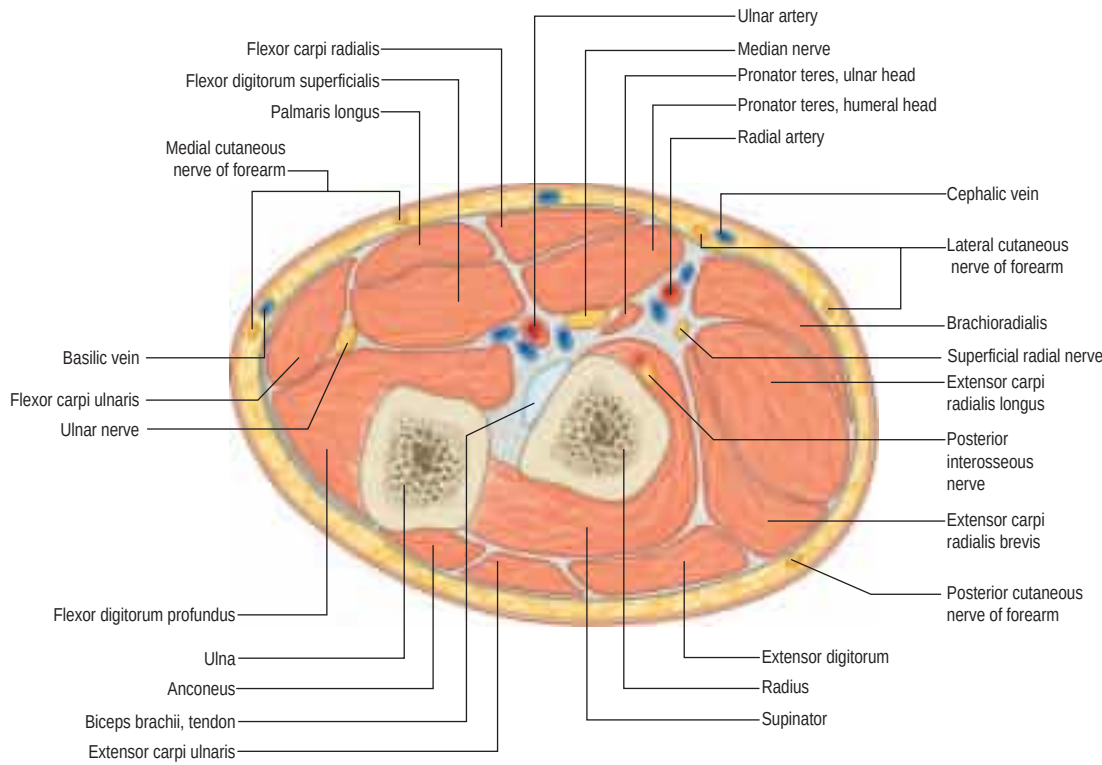
The palmar cutaneous branch starts about 3 cm proximal to the flexor retinaculum. It runs in a small tunnel in the sheath of flexor carpi radialis before piercing the deep fascia, where it divides into lateral branches that supply the thenar skin and connect with the lateral cutaneous nerve of the forearm. Medial branches supply the central palmar skin and connect with the palmar cutaneous branch of the ulnar nerve (see [Figs 46.11, 49.21, 50.35, 50.40A–50.41](#)).



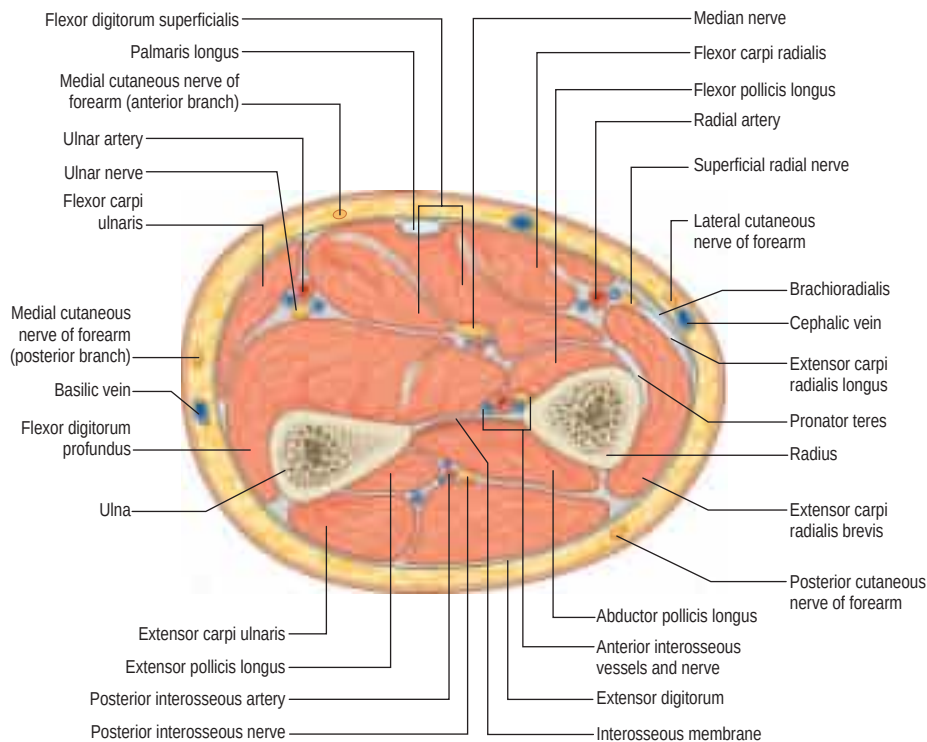
**Fig. 49.30 A–C, Variations in the superficial veins of the left cubital fossa.** (With permission from Waschke J, Paulsen F (eds), *Sobotta Atlas of Human Anatomy*, 15th ed, Elsevier, Urban & Fischer. Copyright 2013.)



The collateral circulation at the elbow is more robust than that at the knee. Alternative pathways include the profunda brachii artery, which accompanies the radial nerve through the lateral intermuscular septum to become the radial collateral artery. The superior ulnar collateral artery accompanies the ulnar nerve. In the adult, it may be as wide as 1.5 mm at its origin; hence its regular use as the artery to a free vascularized ulnar nerve graft. The inferior ulnar collateral artery is regularly seen during medial approaches to the elbow as a vessel that pierces the intermuscular septum to curl around the humerus between triceps and bone. On occasion, there is a substantial artery running with the median nerve. These vessels anastomose with recurrent branches from the ulnar and radial arteries to form a network that may maintain perfusion of the hand and the extensor muscles in cases where the brachial artery has been divided in the lower third of the arm (when there is no associated fracture or dislocation). Collateral circulation will diminish or cease altogether if the ulnar or radial nerve is entrapped within the fracture or compressed by haematoma ([Birch 2011](#), [Blakey et al 2009](#)).



**Fig. 49.31** A transverse section through the left forearm at the level of the radial tuberosity: proximal aspect.



**Fig. 49.32** A transverse section through the middle of the left forearm.

### Muscular branches

Muscular branches are given off near the elbow to all the superficial flexor muscles except flexor carpi ulnaris. The branch to the part of flexor digitorum superficialis that serves the index finger is given off near the mid-forearm and may be derived from the anterior interosseous nerve. The median nerve is palpable in the anterior aspect of the elbow, deep to the bicipital aponeurosis. It is closely related to the brachial artery in the anterior aspect of the elbow, where it courses anterior to brachialis and medial to the artery.

and signs may also be present. More distally, it may become entrapped by a thickened bicipital aponeurosis. The other sites of compression are between the superficial and deep heads of pronator teres, and the proximal fibrous arch of flexor digitorum superficialis.

### Martin-Gruber nerve connection

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### Ulnar nerve

The ulnar nerve passes behind the medial epicondyle in a groove where it is easily palpable (see Figs 49.29, 46.12). It runs through the cubital

### Median nerve compression

The median nerve may be compressed at four sites in the elbow and forearm. The most proximal is the ligament of Struthers, where, if the median nerve is accompanied by the brachial artery, vascular symptoms

Patients essentially present with one of two syndromes. The anterior interosseous nerve syndrome presents as a failure of pinch grip due to weakness of flexor pollicis longus and the radial portion of flexor or digitorum profundus. This reduces the flexion power of the interphalangeal joint of the thumb and the distal interphalangeal joint of the index finger. The pronator syndrome presents with symptoms and signs that mimic carpal tunnel syndrome. Patients present with pain and altered sensation in the radial three and a half fingers of the hand that are aggravated by activity. However, a Tinel's sign will be positive over the proximal forearm, and sensation over the thenar eminence may be diminished, because the palmar cutaneous branch of the nerve arises above the carpal tunnel. The clinical presentation may vary in the presence of a Martin-Gruber anastomosis ([Rodriguez-Niedenführ et al 2002](#)).

Most patients will respond to non-operative treatment but those who fail to do so usually respond well to surgical decompression. Exploration and decompression should address all the potential areas of compression.

A motor communicating branch between the median and ulnar nerves in the forearm occurs in approximately one-quarter of patients. The most common variant has a branch passing from the anterior interosseous to the ulnar nerve. A number of patterns have been described where branches pass between flexors digitorum superficialis and profundus, deep to the ulnar artery, to join the ulnar nerve; such branches may explain the variations in innervation of the intrinsic muscles.



tunnel, between the capsule of the humero-ulnar joint and the overlying arcuate ligament joining the two heads of flexor carpi ulnaris. It gives off articular branches to the elbow as it lies between the medial epicondyle and the olecranon. The ulnar nerve enters the forearm between the two heads of flexor carpi ulnaris, which it supplies by two neurovascular pedicles: one arising just distal to the cubital tunnel and one in the forearm. It is superficial to the posterior and oblique parts of the ulnar collateral ligament.

The ulnar nerve descends on the medial side of the forearm on flexor digitorum profundus (see Fig. 49.32). Proximally, it is covered by flexor carpi ulnaris; its distal half lies lateral to the muscle and is covered only by skin and fasciae. In the upper third of the forearm, the nerve is distant from the ulnar artery but, more distally, it comes to lie close to the medial side of the artery. About 5 cm proximal to the wrist, it gives off the dorsal cutaneous branch. At the wrist, the ulnar nerve passes under the superficial part of the flexor retinaculum (in Guyon's canal) and divides into superficial and deep branches (see Fig. 50.40A).

### Branches in the forearm

#### Muscular branches

There are two main muscular branches of the ulnar nerve in the forearm. They arise near the elbow and supply flexor carpi ulnaris and the medial half of flexor digitorum profundus.

#### Dorsal branch

The dorsal branch of the ulnar nerve continues distally and dorsally, deep to flexor carpi ulnaris. It perforates the deep fascia and descends along the medial side of the back of the wrist and hand. Its further course is described on page 893.

#### Palmar cutaneous branch

The palmar cutaneous branch of the ulnar nerve arises in mid-forearm, descends on the ulnar artery, which it supplies, and then perforates the deep fascia. It communicates with the palmar branch of the median nerve and ends in the palmar skin. It sometimes supplies palmaris brevis. Its further course is described on page 894.

### Ulnar nerve compression

The ulnar nerve may be compressed by the arcade of Struthers at the level of the medial intermuscular septum or at the level of the flexor/pronator aponeurosis as it enters the forearm between the heads of flexor carpi ulnaris. It is most often compressed at the level of the cubital tunnel; this is the most common nerve compression syndrome after that of the median nerve in the carpal tunnel.

### Radial nerve

The radial nerve at the elbow lies in the deep groove between brachialis medially and brachioradialis and extensor carpi radialis longus laterally (see Fig. 49.21). The nerve to brachioradialis arises about three finger's-breadths proximal to the lateral epicondyle; the nerve to extensor carpi radialis longus about one finger's-breadth more distal; and the nerve to extensor carpi radialis brevis about one finger's-breadth above the epicondyle. These branches arise from the main radial nerve as it runs between brachialis and brachioradialis. The nerve may be palpable at this level. The radial nerve divides into the posterior interosseous and superficial radial nerves anterior to the elbow at the level of the tip of the lateral epicondyle. Another branch to extensor carpi radialis brevis is usually given off here. There is some variation in the level at which branches of the radial nerve arise from the main trunk in different individuals. Branches to extensor carpi radialis brevis and supinator may arise from the main trunk of the radial nerve or from the proximal part of the posterior interosseous nerve, but almost invariably above the arcade of Frohse.

### Branches in the forearm

#### Superficial radial nerve

The superficial radial nerve descends from the lateral epicondyle anterolaterally in the proximal two-thirds of the forearm, initially lying on supinator, lateral to the radial artery and behind brachioradialis. In the middle third of the forearm, it lies behind brachioradialis, close to the lateral side of the artery. It runs anterior to pronator teres, the radial head of flexor digitorum superficialis and flexor pollicis longus, and usually leaves the artery 7 cm proximal to the wrist and passes deep to the tendon of brachioradialis. It curves round the lateral side of the radius as it descends, pierces the deep fascia and divides into five, sometimes four, dorsal digital nerves. On the dorsum of the hand, it

usually communicates with the posterior and lateral cutaneous nerves of the forearm.

### Posterior interosseous nerve

The posterior interosseous nerve is the deep terminal branch of the radial nerve (see Figs 46.9, 49.21, 49.29). It reaches the back of the forearm by passing round the lateral aspect of the radius between the two heads of supinator (see Figs 49.25, 49.26). It supplies supinator both before entering the muscle and as it passes through it. As it emerges from supinator posteriorly, the nerve gives off three short branches to extensor digitorum, extensor digiti minimi and extensor carpi ulnaris, and two longer branches: a medial branch to extensor pollicis longus and extensor indicis, and a lateral branch that supplies abductor pollicis longus and extensor pollicis brevis. The nerve at first lies between the superficial and deep extensor muscles, but at the distal border of extensor pollicis brevis, it passes deep to extensor pollicis longus and, diminished to a fine thread, descends on the interosseous membrane to the dorsum of the carpus. Filaments arising from its flattened termination supply the carpal ligaments and articulations. Articular branches from the posterior interosseous nerve supply carpal, distal radio-ulnar and some intercarpal and intermetacarpal joints. Digital branches supply the metacarpophalangeal and proximal interphalangeal joints.

The distal portion of the nerve lies in a separate fascial sheath in the radial, deep, aspect of the fourth dorsal compartment of the extensor retinaculum of the wrist, where it is located deep to extensor digitorum and extensor indicis.

### Radial nerve compression

Radial nerve compression may occur at several sites in the elbow and forearm. The most proximal of these is within the radial tunnel at the level of the radial head, where fibrous bands may be present. More distally, it has been suggested that branches of the radial recurrent artery may compress the posterior interosseous nerve. Other sites of compression are the tendinous edge of the deep surface of extensor carpi radialis brevis and the point at which the nerve passes under the arcade of Frohse and between the two heads of supinator. Rarely, the nerve may be compressed at the level of the distal border of supinator.

### Lateral cutaneous nerve of the forearm

The lateral cutaneous nerve of the forearm is the cutaneous division of the musculocutaneous nerve (p. 831). It lies between biceps brachii and brachialis, inclining from the medial to the lateral side; passes through the cubital fossa lateral to the tendon of biceps brachii (see Figs 48.33, 48.34, 49.31–49.32) and deep to the cephalic vein; and descends along the radial border of the forearm to the wrist. It supplies the skin of the anterolateral surface of the forearm (see Fig. 49.1) and connects with the posterior cutaneous nerve of the forearm and the terminal branch of the radial nerve by branches that pass around its radial border. Its trunk gives rise to a slender recurrent branch that extends along the cephalic vein as far as the middle third of the upper arm, distributing filaments to the skin over the distal third of the anterolateral surface of the upper arm close to the vein. At the wrist joint, the lateral cutaneous nerve of the forearm is anterior to the radial artery. It gives off filaments that pierce the deep fascia and accompany the artery to the dorsum of the carpus. The nerve then passes to the base of the thenar eminence, where it ends in cutaneous rami, some of which connect with the terminal branch of the radial nerve and the palmar cutaneous branch of the median nerve.

### Medial cutaneous nerve of the forearm

The medial cutaneous nerve of the forearm has already divided into anterior and posterior branches before it enters the forearm (see Figs 48.34, 49.12, 49.31–49.32). The larger anterior branch usually passes in front of, or occasionally behind, the median cubital vein, and descends anteromedially in the forearm to supply the skin as far as the wrist. It curves round to the back of the forearm, descending on its medial border to the wrist, supplying the skin (see Fig. 49.1). It connects with the medial cutaneous nerve of the arm, the posterior cutaneous nerve of the forearm, and the dorsal branch of the ulnar nerve.

### Posterior cutaneous nerve of the forearm

The posterior cutaneous nerve of the forearm passes along the dorsum of the forearm to the wrist (see Figs 48.34, 49.31, 49.32). It supplies the central forearm skin along its course and, near its end, joins the dorsal branches of the lateral cutaneous nerve of the forearm.

The cross-sectional area of the cubital tunnel has been shown to narrow by almost 55% in full flexion. Ultrasound studies have suggested that the ratio of the cross-sectional area of radial nerve:cubital tunnel is most strongly linked to reduced conduction velocity (Yoon et al 2007). Combining elbow flexion with direct pressure over the cubital tunnel for 60 seconds will elicit symptoms in most affected patients (Novak et al 1994). Other contributory factors are previous trauma, especially supracondylar and medial epicondyle fractures in childhood, valgus instability of the elbow in athletes involved in throwing sports, and degenerative disease.

The most reliable treatment for patients who fail to respond to non-surgical remedies is a simple release of cubital tunnel. The release should be complete; care should be taken to avoid damage to branches of the medial cutaneous nerve of the forearm that usually cross the line of incision approximately 2 cm above the medial epicondyle and a mean 3.1 cm distal to the medial epicondyle in all patients (Lowe et al 2004). The nerve should not be circumferentially denuded of its retinaculum because this contains its intrinsic blood supply. Transposition of the nerve should be reserved for cases of persistent snapping or of failure of a simple release to achieve symptomatic relief. This is relatively unusual and is said to occur in only 7% of cases.

The ulnar nerve is a mixed nerve and so patients may present with sensory or motor symptoms or a combination of the two, sometimes with additional physical signs of autonomic disturbance. There may be tenderness and a positive Tinel's sign over the nerve at the point of compression. Numbness and paraesthesiae occur early, followed, in more severe cases, by weakness of power grip and, ultimately, atrophy of the lumbricals and interossei with clawing of ring and little fingers.

Patients present with one of two syndromes: the radial tunnel syndrome, in which the presentation is that of pain along the lateral border of the elbow and upper forearm (Roles and Maudsley 1972), or the posterior interosseous nerve syndrome, which presents with motor weakness in the distribution of the nerve particularly affecting the extensors digitorum and extensor carpi ulnaris muscles. Primary treatment is non-operative and involves the avoidance of activities that cause pain, aided by appropriate splintage. In resistant cases, surgical decompression may be necessary. A complete release of all potential areas of compression is essential.

## Nerve injuries at the elbow



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## ANATOMICAL CONSIDERATIONS IN COMMON FOREARM AND ELBOW INJURIES



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**Fig. 49.28** The posterior aspect of the left elbow: superficial structures.

**Fig. 49.33** A type 1 Monteggia fracture with anterior dislocation of the radial head.

**Fig. 49.34** A Galeazzi fracture-dislocation.

**Fig. 49.35** Elbow dislocation.

The elbow is a notorious site for nerve injuries, especially in children. The close proximity of all the major nerve trunks to bone makes them vulnerable. The ulnar nerve is particularly at risk in elbow dislocation, especially when the medial epicondyle is displaced into the joint. All three nerves are at risk of displacement into a dislocated joint. The posterior interosseous and radial nerves are particularly at risk from a Monteggia fracture-dislocation. Any nerve may become entrapped within a fracture, e.g. the median nerve is often accompanied by the brachial artery in cases of supracondylar fracture. Pain that persists, or worsens, after reduction of a fracture suggests impending critical ischaemia, continuing irritation of the nerve, or both. The nerves are at risk during operations or other medical interventions, particularly so when normal anatomical planes are distorted by haematoma, fibrosis or bone fragments.

Fractures of the shaft of the radius and ulna are commonly displaced, reflecting not only the violence required to cause such injury but also the pull of the forearm muscles that may accentuate displacement of the fracture fragments. Open fractures or penetrating injuries to the forearm may be associated with nerve and vessel injury. Compartment syndrome can occur. Monitoring of compartment pressures and careful neurological examination are required.

### Fracture-dislocations of the forearm

Monteggia, Galeazzi and Essex–Lopresti injuries are common combinations of high-energy forearm fractures associated with accompanying soft tissue injury.

#### Monteggia fracture-dislocation

A Monteggia fracture-dislocation injury is a fracture of the proximal ulna with dislocation of the radial head; the interosseous membrane and the triangular fibrocartilage complex remain largely intact, which means that anatomical reduction of the ulnar fracture will usually restore congruity of the radiocapitellar and proximal radio-ulnar joints. The most common type of Monteggia fracture, type 1 (anterior dislocation), occurs when falling on to an outstretched arm hyperextends the elbow (Fig. 49.33). The forearm is in neutral to mid-pronation, biceps brachii resists the hyperextension, the anular and quadrate ligaments rupture under the bending force, and the radial head dislocates. Compressive load then transfers to the ulnar diaphysis, which fails in tension in an oblique fracture. Type II (posterior dislocation) involves longitudinal force directed up the forearm with the elbow in a flexed position. The ulnar cortex fails first before rupture of the ulnar collateral ligament complex and posterior dislocation of the radial head. Four subtypes of this injury pattern have been identified, based on the location and force direction through the ulnar fracture (Jupiter et al 1991). Type III Monteggia injuries result in lateral or anterolateral dislocation of the radial head, associated with a metaphyseal fracture of the proximal ulna. Type IV Monteggia injuries result in an anterior dislocation of the radial head and a fracture through the proximal third of both the radius and the ulna. The mechanism of injury is similar to that seen in type I injuries but the anteriorly directed force of the distal ulnar fragment, combined with the biceps resistance force, causes a fracture through the proximal radius (Waters 2010).

#### Galeazzi fracture-dislocation

A Galeazzi injury is a fracture at the junction of the middle and distal thirds of the radius, associated with disruption of the distal radio-ulnar joint (Fig. 49.34). The injury occurs when the forearm is axially loaded and fully pronated. The radius fractures first, force transmission continues along the interosseous membrane to the distal ulna, the triangular fibrocartilage complex is disrupted and the ulnar head dislocates. The ulnar styloid can also be fractured in this injury (Galeazzi 1934).

#### Essex–Lopresti injury

Essex–Lopresti injury is a radio-ulnar dissociation injury, usually caused by a fall on to an outstretched hand that involves a fracture of the radial head, force transmission between the radius and ulna, and disruption of the interosseous membrane and the distal radio-ulnar joint. The radius migrates proximally relative to the ulna (Chow and Leung 2010).



Fig. 49.33 A type 1 Monteggia fracture with anterior dislocation of the radial head (arrow).



Fig. 49.34 A Galeazzi fracture-dislocation.



***Traumatic elbow instability***

Posterolateral rotatory pattern instability and dislocation of the elbow, with or without associated fracture of the radial head and coronoid, are the three general patterns of traumatic elbow instability (Ring 2010). Posterolateral rotatory pattern instability is an injury of the capsule and ligaments that progresses from lateral to medial with a posterolateral rotatory mechanism (O'Driscoll et al 1992). It usually occurs as the result of a fall on to an outstretched hand that creates a valgus, axial posterolateral rotatory force. The ulna and forearm supinate away from the distal humerus and a posterior dislocation of the elbow occurs (Fig. 49.35). As the elbow dislocates posteriorly, the radial head and coronoid impact on the distal humerus, where either can fracture; the anterior band of the ulnar collateral ligament is the last structure to be disrupted. Varus posteromedial rotational pattern instability and anteromedial coronoid facet fractures occur when a fall on to an outstretched hand produces a varus, axial posteromedial force through the elbow. The anteromedial facet of the coronoid is fractured and the lateral ligament complex may be injured or the olecranon may be fractured. Anterior olecranon fracture-dislocations occur as a result of direct violence to the flexed elbow. The mechanism of posterior olecranon fracture dislocation is less certain; it occurs more commonly in osteopaenic bone.



**Fig. 49.35** Elbow dislocation.

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